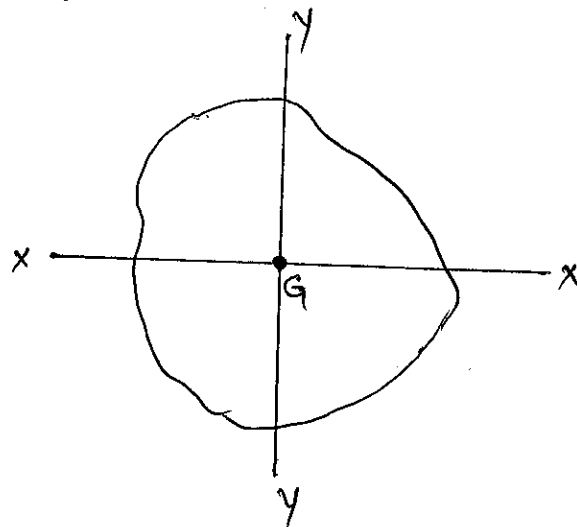


MODULE - 3: CENTROIDCentre of Gravity [CG or G]

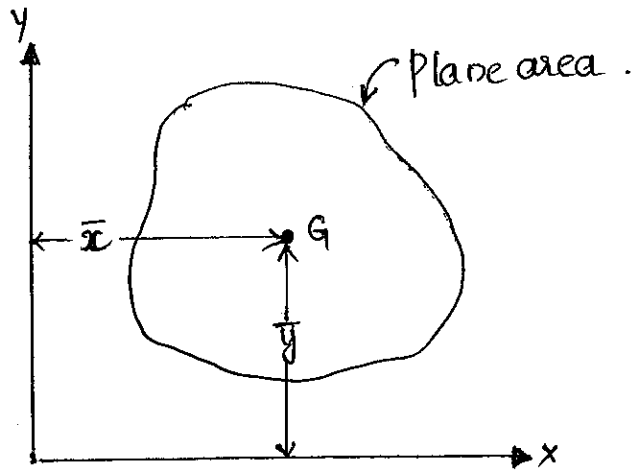
Centre of gravity of a body is a point through which the whole weight of the body acts. It is represented by CG (or) G. There is only one centre of gravity for any position of body. It refers to the body with mass & weight.

Centroid.

Centroid of an area is a point at which the whole area of the plane figure is assumed to be concentrated. It is represented by CG (or) G. The centroid & centre of gravity lies in the same point 'G' but refers to plane area of bodies.

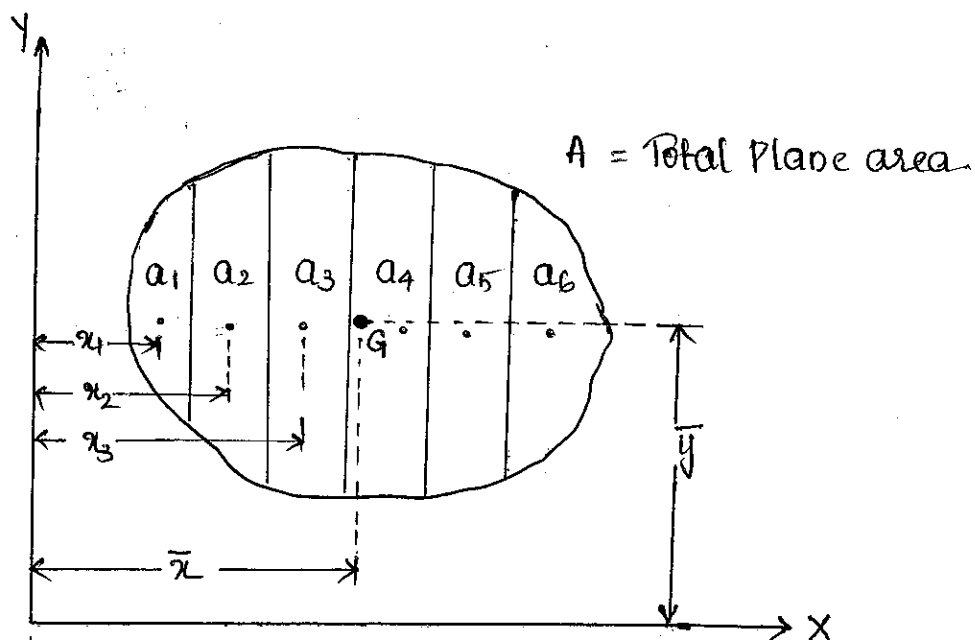
The difference between centroid & centre of gravity is that centre of gravity applies to the bodies with mass & weight, while the centroid refers to the plane bodies, areas & volumes of the body.

Location of Centroid of Plane areas.



The position of centroid of a plane area should be specified (or) calculated w.r.t reference axis i.e., x & y axis. The location of centroid 'G' from the vertical reference axis (or) y -axis is denoted as $\bar{x}$ & distance of centroid 'G' from a horizontal reference axis (or) x -axis is denoted as $\bar{y}$.

Method of moments to locate the Centroid of Plane areas.



Let us consider a plane area (A) lying in the xy plane. Let 'G' be the centroid of the plane area.

It is required to locate the position of centroid 'G' wrt x & y axis.

Let us divide the area A into smaller elemental areas a_1, a_2, a_3, a_4 ----- as shown in fig.

Let g_1, g_2, g_3, g_4 ----- be the centroid of elemental areas a_1, a_2, a_3, a_4 ----- etc.

Let x_1, x_2, x_3 ----- be the distance of centroids g_1, g_2, g_3 ----- from y-axis i.e., $= A\bar{x}$ ——— (1)

The sum of the moments of the elemental areas about y-axis is $a_1x_1 + a_2x_2 + a_3x_3 +$ ----- (2)

Equating (1) & (2)

$$A\bar{x} = a_1x_1 + a_2x_2 + a_3x_3 + \dots$$

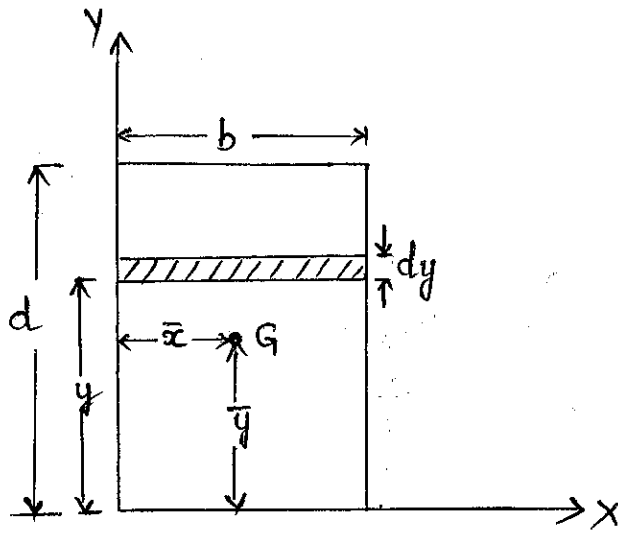
$$\bar{x} = \frac{a_1x_1 + a_2x_2 + a_3x_3 + \dots}{A}$$

$$\boxed{\bar{x} = \frac{\sum ax}{A}}$$

(or)

$$\boxed{\bar{x} = \frac{\int x \cdot dA}{A}}$$

Location of Centroid of a rectangle from the first principle method (Method of moments)



Let us consider a rectangle of breadth 'b' & depth 'd'.

Let G be the centroid of the rectangle. Let us consider x & y axis as shown in fig.

Let us consider an elemental area dA of breadth b & depth dy lying at a distance of 'y' from the x-axis.

$$\text{w.k.T, } \bar{y} = \frac{\int_0^d y \cdot dA}{A} \quad \text{--- (1)}$$

Also, $A = bd$.

Elemental area, $dA = b \cdot dy$.

On substitution in eqn (1)

$$\bar{y} = \frac{\int_0^d y \cdot (b \cdot dy)}{b \cdot d}$$

$$\bar{y} = \frac{1}{d} \int_0^d y \cdot dy$$

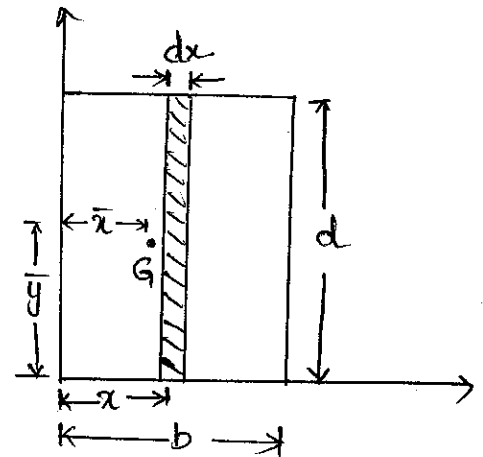
$$\bar{y} = \frac{1}{d} \left[\frac{y^2}{2} \right]_0^d$$

$$\bar{y} = \frac{1}{d} \cdot \left(\frac{d^2}{2} \right)$$

$$\boxed{\bar{y} = \frac{d}{2}}$$

Similarly, $\bar{x} = \frac{\int_0^b x \cdot dA}{A} \quad \text{--- (2)}$

Consider an elemental area 'dA' of depth 'd' & breadth 'dx' at a distance of 'x' from y-axis.



Area, $A = bd$.

Elemental area, $dA = dx \cdot d$.

On substitution in eqn (2)

$$\bar{x} = \frac{\int_0^b x \cdot (dx \cdot d)}{b \cdot d}$$

$$\bar{x} = \frac{1}{b} \int_0^b x \cdot dx$$

$$\bar{x} = \frac{1}{b} \left[\frac{x^2}{2} \right]_0^b$$

$$\bar{x} = \frac{1}{b} \left[\frac{b^2}{2} \right]$$

$$\boxed{\bar{x} = \frac{b}{2}}$$

Centroid of a triangle

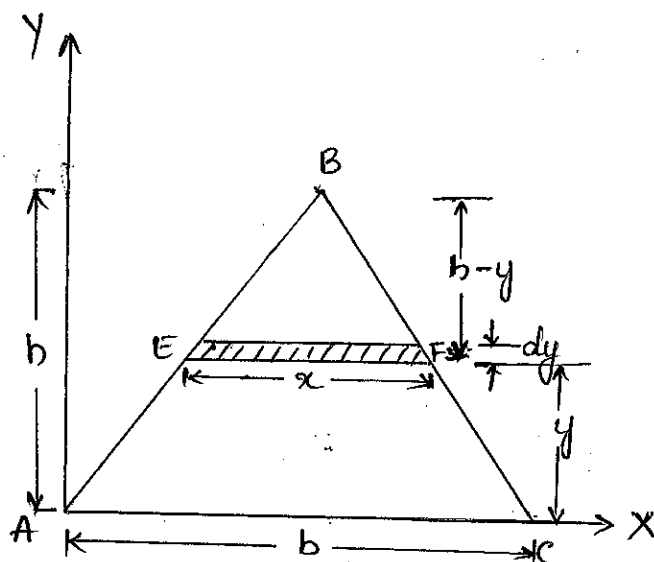


Fig. shows a triangle of base 'b' & height 'h' w.r.t x & y-axis. Let us consider an elemental strip of base 'x' & thickness 'dy' at a distance 'y' from the x-axis.

Let $A \rightarrow$ Total area of triangle.

$dA \rightarrow$ area of elemental strip EF.

Area of elemental strip is approximately a rectangle.

$$\therefore dA = x \cdot dy. \quad \text{--- (1)}$$

Using similar triangles,

$$\frac{x}{h-y} = \frac{b}{h}$$

$$x = \frac{b(h-y)}{h} \quad \text{--- (2)}$$

From (1) & (2)

$$dA = \frac{b \cdot (h-y)}{h} \cdot dy$$

On integration,

$$A = \int_0^h \frac{b(h-y)}{h} \cdot dy$$

$$A = \int_0^h \left(b - \frac{by}{h} \right) \cdot dy.$$

$$A = \left[b \cdot y - \frac{b \cdot y^2}{2h} \right]_0^h$$

$$A = bh - \frac{b \cdot h^2}{2h}$$

$$A = bh - \frac{bh}{2}$$

$$\boxed{A = \frac{1}{2}bh}$$

W.K.P, $\bar{y} = \frac{\text{Moment of area}}{\text{Total area}}$

$$\text{i.e., } \bar{y} = \frac{\int y \cdot dA}{A}$$

Substituting for dA & A

$$\bar{y} = \frac{\int_0^h y \cdot \frac{b}{h}(h-y) \cdot dy}{\frac{1}{2} \cdot b \cdot h}$$

$$\bar{y} = \frac{\frac{b}{h} \left[\frac{h \cdot y^2}{2} - \frac{y^3}{3} \right]_0^h}{\frac{1}{2} \cdot b \cdot h}$$

$$\bar{y} = \frac{\frac{b}{h} \left[\frac{h^3}{2} - \frac{h^3}{3} \right]}{\frac{1}{2} \cdot b \cdot h}$$

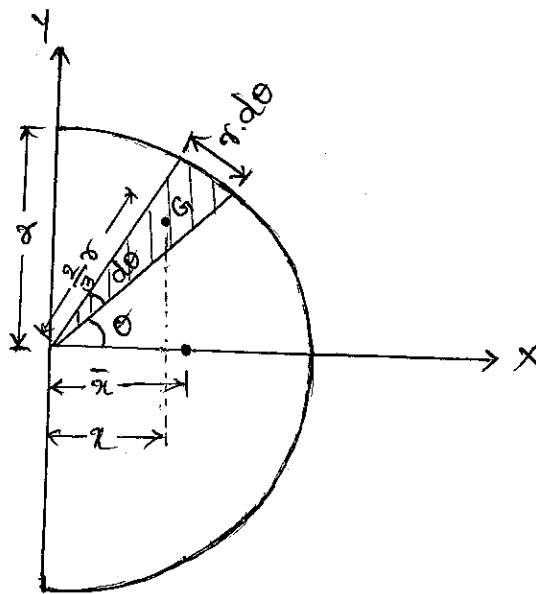
$$\bar{y} = \frac{\frac{b}{h} \left(\frac{h^3}{6} \right)}{\frac{1}{2}bh.}$$

$$\bar{y} = \frac{\frac{bh^3}{6h}}{\frac{bh}{2}}$$

$$\boxed{\bar{y} = \frac{h}{3}} \text{ from base}$$

$$\boxed{\bar{y} = \frac{2h}{3}} \text{ from the apex (or) tip.}$$

Centroid of a semicircle of radius 'r'



Let us consider a semicircle of radius 'r' as shown in fig. Due to symmetry, centroid must lie on x-axis

$$\boxed{\bar{y} = 0} \quad (\text{or}) \quad \boxed{\bar{y} = r(\theta) d/2}$$

Consider an elemental area of inscribed angle 'dθ' as shown in fig. As 'dθ' is very small, the area is a triangle having its centroid at a distance $\frac{2r}{3}$ from O.

The area of the triangular element,

$$dA = \frac{1}{2} \times r \times r d\theta$$

$$dA = \frac{1}{2} \times r^2 \times d\theta$$

$$A = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1}{2} r^2 \cdot d\theta$$

$$A = \frac{r^2}{2} \left[\theta \right]_{-\frac{\pi}{2}}^{\frac{\pi}{2}}$$

$$A = \frac{r^2}{2} \left[\frac{\pi}{2} - \left(-\frac{\pi}{2} \right) \right]$$

$$\boxed{A = \frac{\pi r^2}{2}}$$

The x-coordinate of centroid of the triangular elemental area is $\frac{2}{3} r \cos \theta = x$.

$$\bar{x} = \frac{\text{moment of area}}{\text{total area}}$$

$$= \frac{\int x \cdot dA}{A}$$

$$= \frac{\int_{-\pi/2}^{\pi/2} \left(\frac{2}{3} r \cos \theta \right) \left(\frac{1}{2} r^2 d\theta \right)}{\frac{\pi r^2}{2}}$$

$$= \frac{\frac{r^3}{3}}{\frac{\pi r^2}{2}} \int_{-\pi/2}^{\pi/2} \cos \theta \cdot d\theta$$

$$\bar{x} = \frac{2r}{3\pi} \left[\sin \theta \right]_{-\pi/2}^{\pi/2}$$

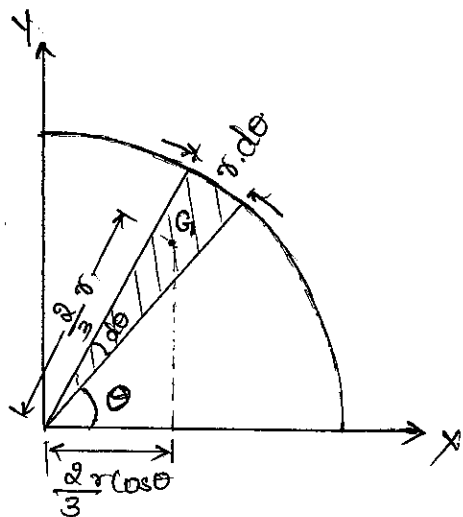
$$\bar{x} = \frac{2r}{3\pi} \left[\sin \pi/2 - \sin(-\pi/2) \right]$$

$$\bar{x} = \frac{2r}{3\pi} [1 - (-1)]$$

$$\bar{x} = \frac{2r}{3\pi} [2]$$

$$\boxed{\bar{x} = \frac{4r}{3\pi}}$$

Centroid of a quarter/quadrantal circle of radius 'r'



Consider a quarter circle of radius 'r' as shown in fig. Consider an elementary area of inscribed angle 'dθ' as shown. As 'dθ' is very small the area is a triangle having its centroid at a distance $\frac{2r}{3}$ from 'O'.

The area of triangular element is,

$$dA = \frac{1}{2} \cdot r \times r \cdot d\theta$$

$$dA = \frac{1}{2} \cdot r^2 \cdot d\theta$$

$$A = \int_0^{\pi/2} \frac{1}{2} r^2 \cdot d\theta$$

$$A = \frac{1}{2} r^2 \int_0^{\pi/2} d\theta$$

$$A = \frac{1}{2} r^2 [\theta]_0^{\pi/2}$$

$$A = \frac{1}{2} r^2 \left[\frac{\pi}{2} \right]$$

$$\boxed{A = \frac{\pi r^2}{4}}$$

The x-coordinate of centroid of the triangular elementary area, $x = \frac{2}{3} r \cos \theta$

$$\bar{x} = \frac{\text{moment of area}}{\text{total area}}$$

$$\bar{x} = \frac{\int x \cdot dA}{A}$$

$$\bar{x} = \frac{\int_0^{\pi/2} \left(\frac{2}{3} r \cos \theta \right) \left(\frac{1}{2} r^2 \cdot d\theta \right)}{\frac{\pi r^2}{4}}$$

$$\bar{x} = \frac{r^3/3 \int_0^{\pi/2} \cos \theta \cdot d\theta}{\frac{\pi r^2}{4}}$$

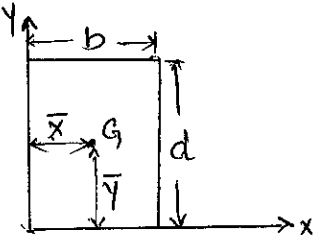
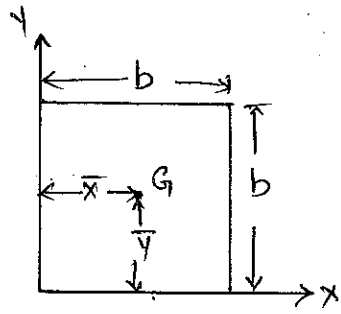
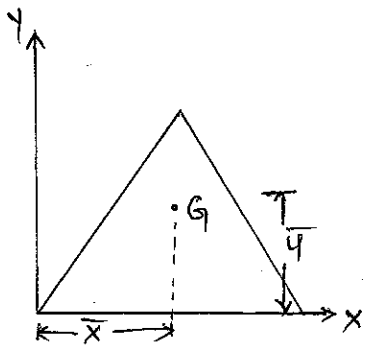
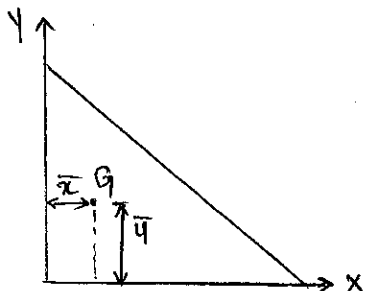
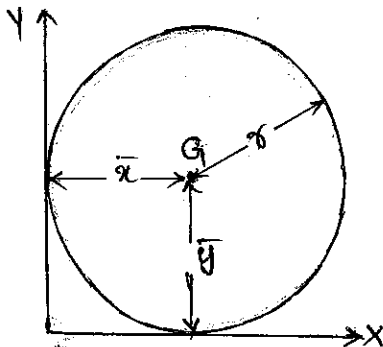
$$\bar{x} = \frac{4r}{3\pi} [\sin \theta]_0^{\pi/2}$$

$$\bar{x} = \frac{4r}{3\pi} \left[\sin \frac{\pi}{2} - \sin 0 \right]$$

$$\boxed{\bar{x} = \frac{4r}{3\pi}}$$

Similarly, $\boxed{\bar{y} = \frac{4r}{3\pi}}$

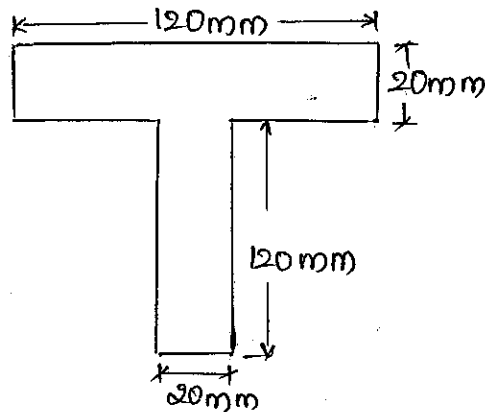
Location of Centroids of various geometrical figures

Sl. No.	Description	Figure	Area (A)	$\bar{x}$	$\bar{y}$
1.	Rectangle		$b \times d$	$\frac{b}{2}$	$\frac{d}{2}$
2.	Square		b^2	$\frac{b}{2}$	$\frac{b}{2}$
3.	Triangle		$\frac{1}{2} \times b \times h$	$\frac{b}{2}$	$\frac{h}{3}$ from base $\frac{2h}{3}$ from apex
4.	Right angle triangle		$\frac{1}{2} \times b \times h$	$\frac{b}{3}$ from origin	$\frac{h}{3}$ from base
5.	Circle		πr^2	r	r

Sl. no.	Description	Figure	Area (A)	$\bar{x}$	$\bar{y}$
6.	Semicircle		$\frac{\pi r^2}{2}$	$\frac{4r}{3\pi}$	$r(0.4244)$
			$\frac{\pi r^2}{2}$	$-\frac{4r}{3\pi}$	$r(0.4244)$
			$\frac{\pi r^2}{2}$	$r(0.4244)$	$\frac{4r}{3\pi}$
			$\frac{\pi r^2}{2}$	$r(0.4244)$	$-\frac{4r}{3\pi}$
			$\frac{\pi r^2}{4}$	$\frac{4r}{3\pi}$	$\frac{4r}{3\pi}$
7.	Quadrantal (or) quarter circle		$\frac{\pi r^2}{4}$	$\frac{4r}{3\pi}$	$\frac{4r}{3\pi}$

Problems.

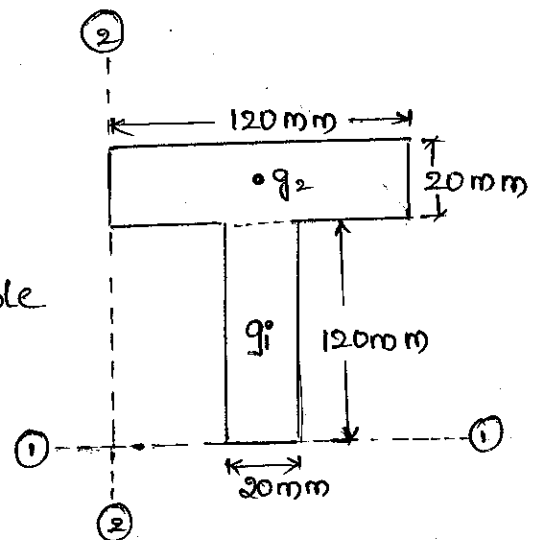
1- locate the centroid of the T-section shown in fig.



Solⁿ:

Let us consider the reference as (1)-(1) & (2)-(2) as shown in fig.

Divide the given figure into simple figure. The fig. is symmetrical about y-y axis.



$\therefore$ required to find $\bar{y} = \frac{\sum ay}{\sum a}$

Component	Area (a) mm ²	Distance of centroid of component from (1)-(1), (y) mm	Moment of area about (1)-(1), (ay) mm ³
g_1	$b \times d$ $20 \times 120 = 2400$	$\frac{d}{2} = \frac{120}{2} = 60$	144000
g_2	$b \times d$ $120 \times 20 = 2400$	$120 + \frac{d}{2} = 120 + \frac{20}{2} = 130$	312000

$$\sum a = 4800$$

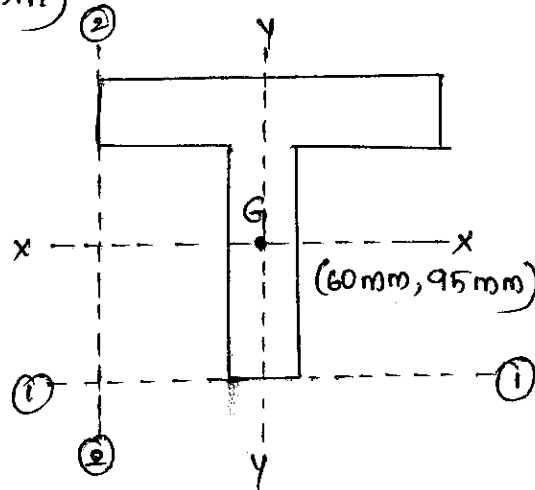
$$\sum ay = 456000$$

$$\therefore \bar{y} = \frac{456000}{4800}$$

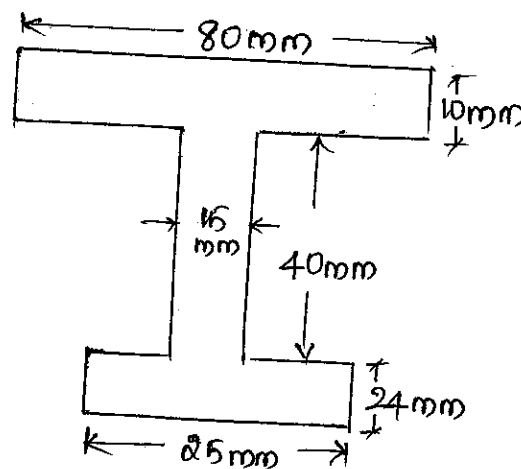
$$\bar{y} = 95 \text{ mm}$$

$$\bar{X} = \frac{120}{2} = \underline{\underline{60 \text{ mm}}}$$

The co.ordinates of centroid of given T-section are (60 mm, 95 mm) as shown below-



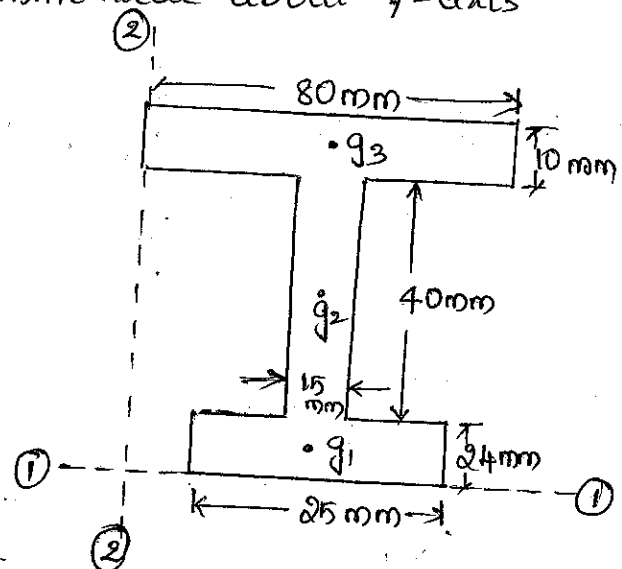
2. Locate the centroid of the T-section shown in fig.



Sol: Consider the reference axis 1-1 & 2-2 as shown in fig. The given fig. is symmetrical about y-axis

$$\bar{X} = \frac{80}{2} = 40 \text{ mm}$$

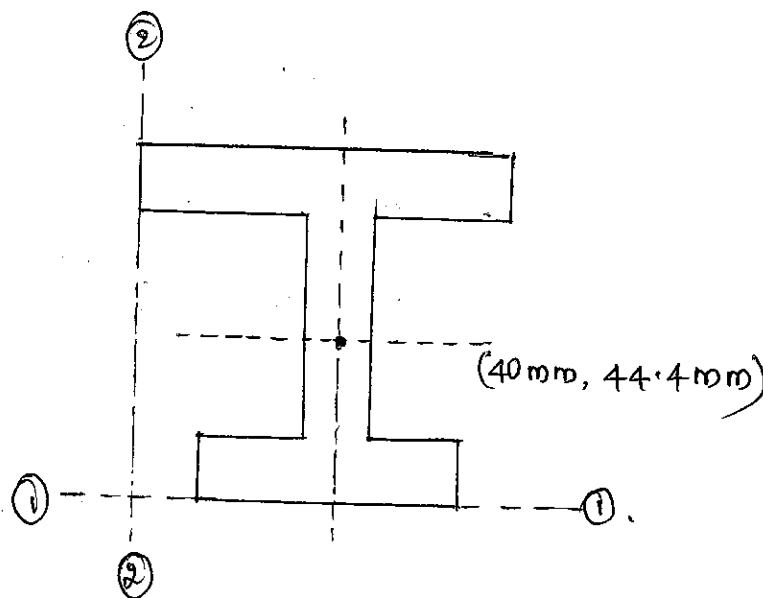
$$\bar{Y} = \frac{\sum ay}{\sum a}$$



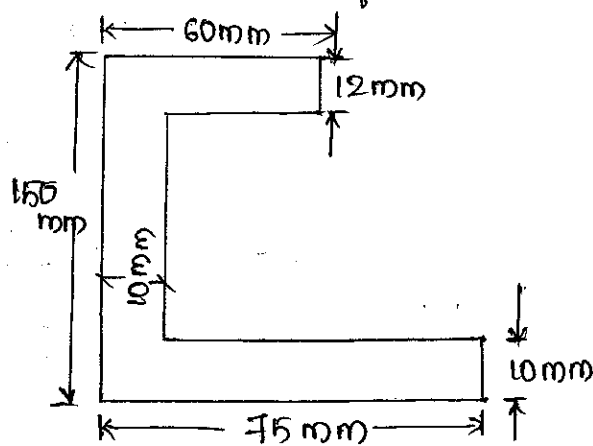
Component	Area (a) mm ²	Distance of centroid of component from (1)-(1) y, mm	Moment of area about (1)-(1), ay (mm ³)
g ₁	b × d 25 × 24 = 600	$\frac{24}{2} = 12$	7200
g ₂	b × d 40 × 15 = 600	$24 + \frac{40}{2} = 44$	26400
g ₃	b × d 80 × 10 = 800	$24 + 40 + \frac{10}{2} = 69$	55200
$\Sigma a = 2000$			$\Sigma ay = 88,800$

$$\bar{y} = \frac{88,800}{200} = \underline{\underline{44.4 \text{ mm}}}$$

Co-ordinates of centroid G are (40 mm, 44.4 mm) as shown in fig.



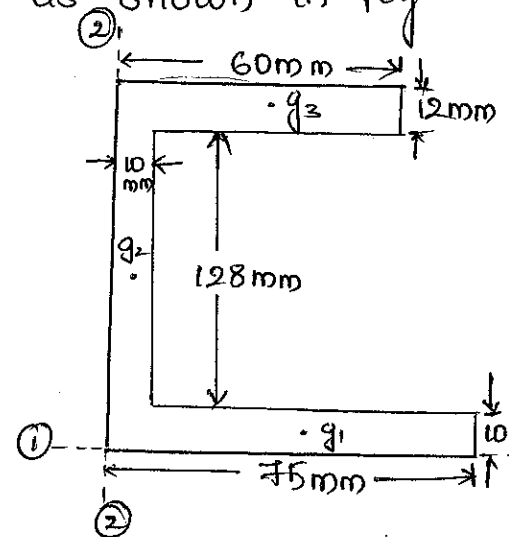
3. Locate the centroid of the C-section shown in fig.



Solⁿ: Consider the reference axis as shown in fig

$$\bar{y} = \frac{\sum ay}{\sum a}$$

$$\bar{x} = \frac{\sum ax}{\sum a}$$



Component	Area (a) mm ²	y - mm	x - mm	ay mm ³	ax mm ³
g ₁	b × d 75 × 10 = 750	$\frac{10}{2} = 5$	$\frac{75}{2} = 37.5$	3750	28125
g ₂	128 × 10 = 1280	$10 + \frac{128}{2} = 74$	$\frac{10}{2} = 5$	94720	6400
g ₃	60 × 12 = 720	$10 + 128 + \frac{12}{2} = 144$	$\frac{60}{2} = 30$	103680	21600

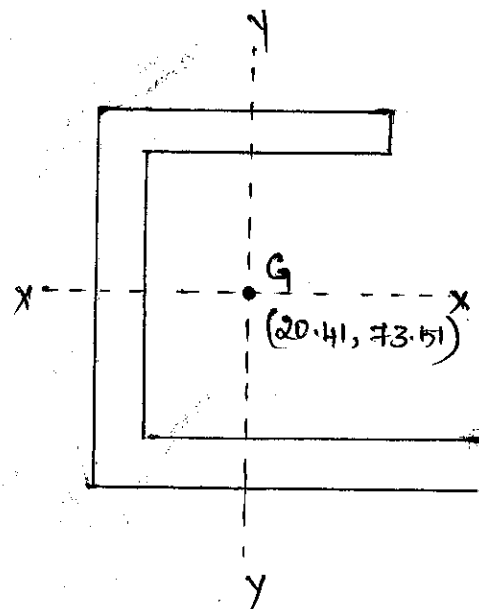
$$\sum a = 2750$$

$$\sum ay = 202150$$

$$\sum ax = 56125$$

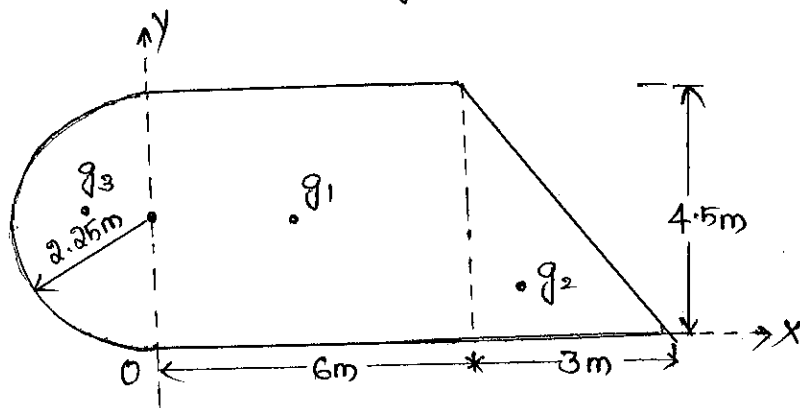
$$\bar{y} = \frac{202150}{2750} = 73.51 \text{ mm}$$

$$\bar{x} = \frac{56125}{2750} = 20.41 \text{ mm}$$



location of CG.

4. Locate the centroid of the area shown in figure w.r.t the axis shown in fig.



Solⁿ: The given fig. is unsymmetrical.
 $\therefore$ We need to find both $\bar{x}$ & $\bar{y}$

$$\bar{y} = \frac{\sum ay.}{\sum a} \quad \& \quad \bar{x} = \frac{\sum ax.}{\sum a.}$$

Component	Area(a) m ²	Distance y - m	Distance x - m	ay in m ³	ax in m ³
g ₁ (rectangle)	bxd 6 × 4.5 = 27	$\frac{4.5}{2} = 2.25$	$\frac{6}{2} = 3$	60.75	81
g ₂ (triangle)	$\frac{1}{2}bh$ $\frac{1}{2} \times 3 \times 4.5$ = 6.75	$\frac{h}{3} = \frac{4.5}{3}$ = 1.5	$6 + \frac{3}{3} = 7$	10.13	47.25
g ₃ (semicircle)	$\frac{\pi r^2}{2} = \frac{\pi (2.25)^2}{2}$ = 7.96	r = 2.25	$\frac{-4r}{3\pi} = \frac{-4(2.25)}{3\pi}$ = -0.954	17.89	-7.58

$$\sum a = 41.7$$

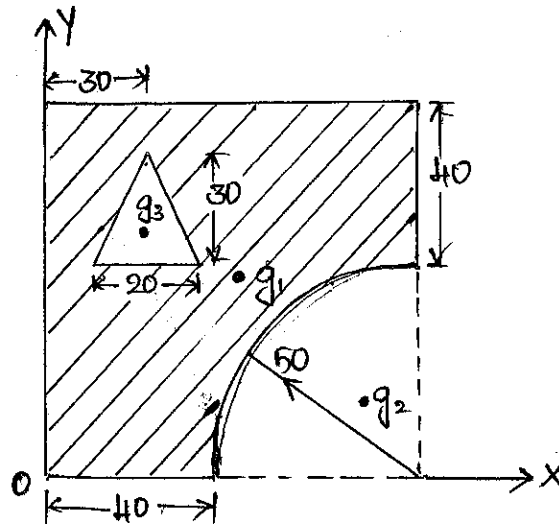
$$\sum ay = 88.77 \quad \sum ax = 120.67$$

$$\bar{y} = \frac{88.77}{41.7} = \underline{\underline{2.13 \text{ m.}}}$$

$$\bar{x} = \frac{120.67}{41.7} = \underline{\underline{2.89 \text{ m}}}$$

The co-ordinates of the given figure are (2.89, 2.13) m.

5. Locate the centroid of the shaded area shown in fig.



(All dimensions are in mm)

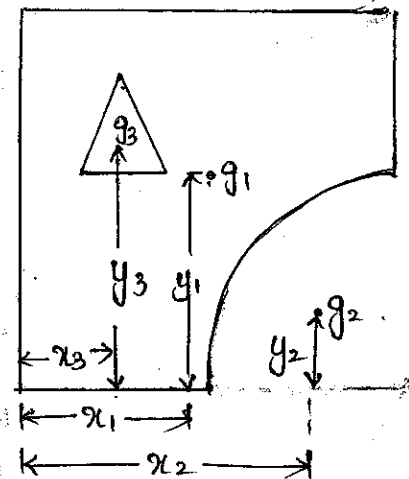
Solⁿ:-

Consider the reference axis as shown in fig.

Fig. is unsymmetrical.

$\therefore$ Find $\bar{x}$ & $\bar{y}$

$$\bar{y} = \frac{\sum ay}{\sum a} \quad \& \quad \bar{x} = \frac{\sum ax}{\sum a}$$



Component	Area (a) mm ²	y - mm	x - mm	ay - mm ³	ax = mm ³
g ₁ (Square)	90 × 90 = 8100	90/2 = 45	90/2 = 45	364500	364500
<u>Deductions</u> g ₂ (Quarter circle)	$\frac{\pi r^2}{4} = \frac{\pi (50)^2}{4}$ = 1963.5	$\frac{4r}{3\pi} = \frac{4 \times 50}{3\pi}$ = 21.2	$90 - \frac{4r}{3\pi}$ = 68.8	41626.2	135088.8
g ₃ (triangle)	$\frac{1}{2} \times 20 \times 30$ = 300	$50 + \frac{30}{3} = 60$	30	18000	9000
	$\sum a = 5836.5$			$\sum ay =$ 304873.8	$\sum ax =$ 220411.2

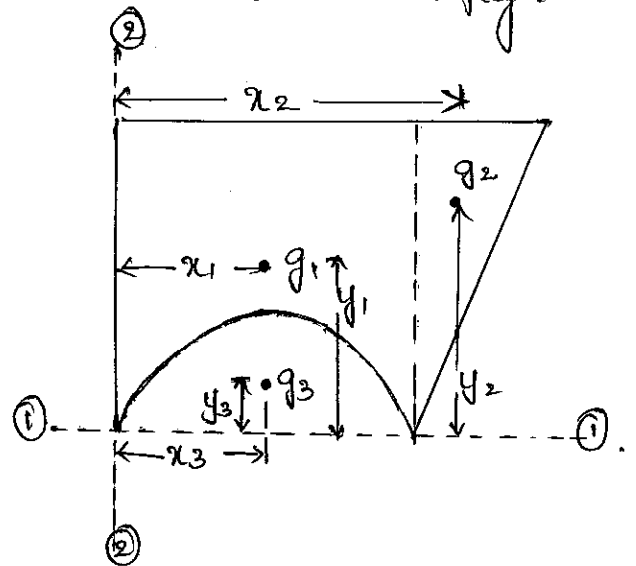
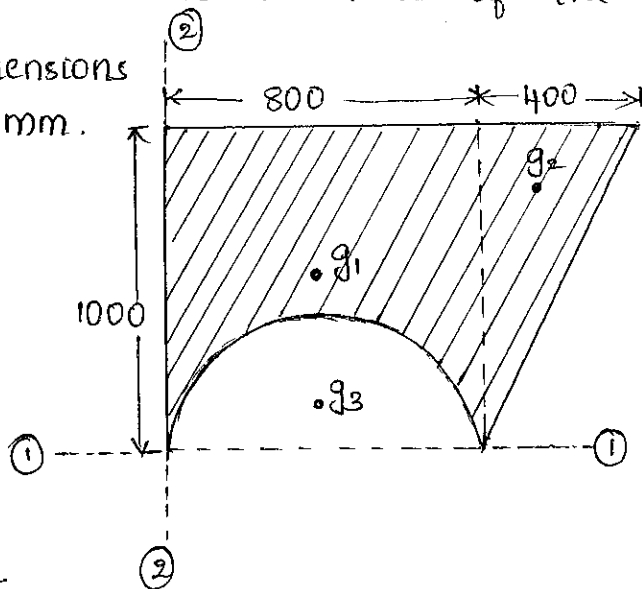
$$\bar{y} = \frac{304873.8}{5836.5} = \underline{\underline{52.24 \text{ mm}}}$$

$$\bar{x} = \frac{220411.2}{5836.5} = \underline{\underline{37.76 \text{ mm}}}$$

The co-ordinates of the given fig is (37.76 mm, 52.24 mm)

6. Locate the centroid of the shaded area shown in fig.

All dimensions are in mm.



Solⁿ:

Consider the reference axis as shown in fig. The fig is unsymmetrical.
 $\therefore$ find $\bar{x}$ & $\bar{y}$, $\bar{y} = \frac{\sum ay}{\sum a}$ & $\bar{x} = \frac{\sum ax}{\sum a}$

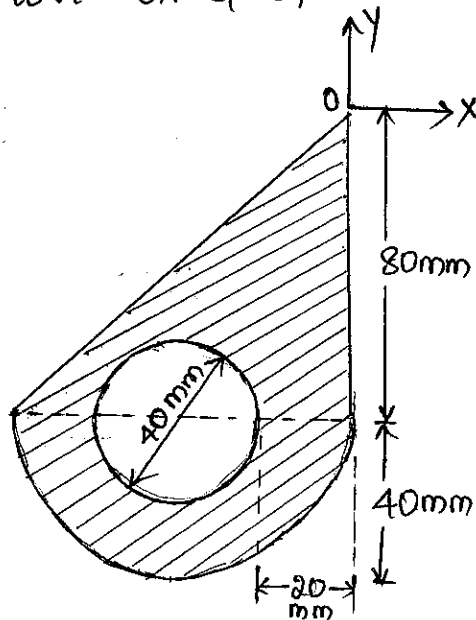
Component	Area (a) mm ²	y - mm	x - mm	ay - mm ³	ax - mm ³
g ₁ (Rectangle)	800 × 1000 = 8 × 10 ⁵	$\frac{1000}{2} = 500$	$\frac{800}{2} = 400$	4 × 10 ⁸	3.2 × 10 ⁸
g ₂ (Triangle)	$\frac{1}{2} \times 400 \times 1000$ = 2 × 10 ⁵	$\frac{2}{3} (1000)$ = 666.67	$800 + \frac{1}{3} (400)$ = 933.33	1.33 × 10 ⁸	1.867 × 10 ⁸
g ₃ (Semi-circle)	$\frac{\pi r^2}{2} = \frac{\pi (400)^2}{2}$ = 2.51 × 10 ⁵	$\frac{4r}{3\pi} = \frac{4(400)}{3\pi}$ = 169.6	$\frac{800}{2} = 400$	0.426 × 10 ⁸	1.005 × 10 ⁸
	$\sum a = 7.49 \times 10^5$			$\sum ay = 4.904 \times 10^8$	$\sum ax = 4.062 \times 10^8$

$$\bar{y} = \frac{4.904 \times 10^8}{7.49 \times 10^5} = \underline{\underline{654.53 \text{ mm}}}$$

$$\bar{x} = \frac{4.062 \times 10^8}{7.49 \times 10^5} = \underline{\underline{542.15 \text{ mm}}}$$

$\therefore$ The co-ordinates of the given fig. is $(542.15 \text{ mm}, 654.53 \text{ mm})$

7. Determine the centroid of the shaded area shown in fig. wrt ox & oy.



Solⁿ:

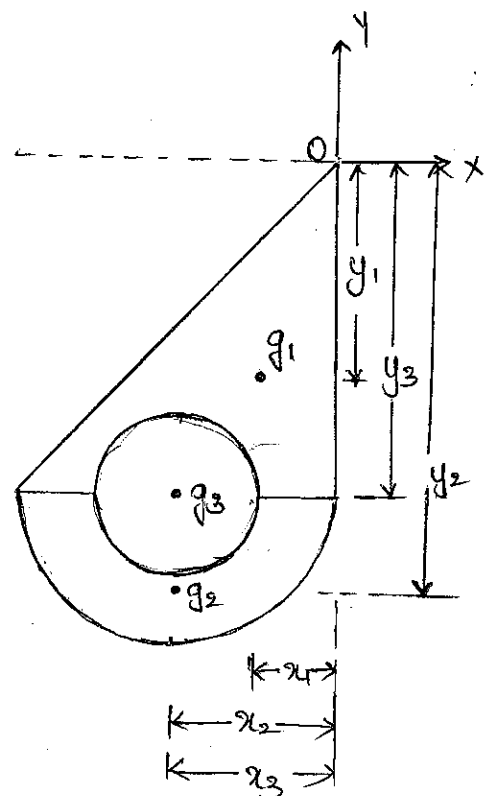
Consider the reference axis ox & oy as shown in fig.

The fig. is unsymmetrical.

$\therefore$ find $\bar{x}$ & $\bar{y}$

$$\bar{y} = \frac{\sum ay}{\sum a}$$

$$\bar{x} = \frac{\sum ax}{\sum a}$$



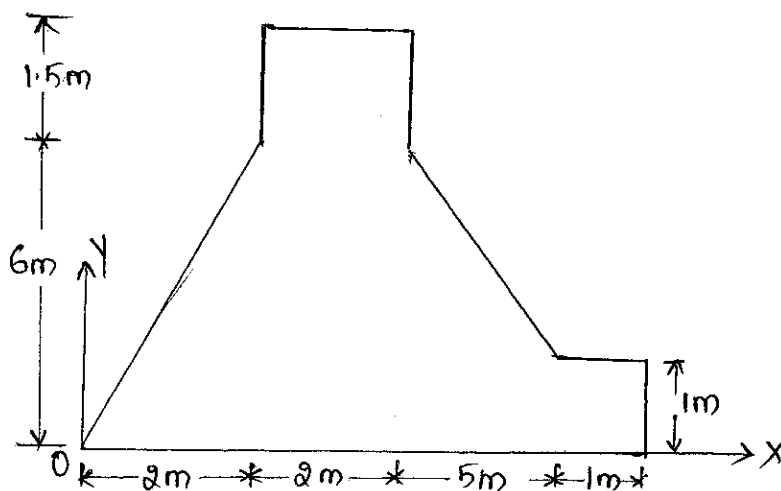
Component	Area (a) mm^2	y - mm	x - mm	ay - mm^3	ax - mm^3
g_1 (Triangle)	$\frac{1}{2} \times b \times h$ $\frac{1}{2} \times 80 \times 80$ $= 3200$	$-\frac{2}{3} \times 80 = -53.33$	$-\left(\frac{80}{3}\right)$ $= -26.67$	-170.65×10^3	-85.34×10^3
g_2 (Semicircle)	$\frac{\pi r^2}{2} = \frac{\pi (40)^2}{2}$ $= 2513.27$	$-\left(\frac{80 + 40}{3\pi}\right)$ $= -96.98$	-40	-243.74×10^3	-100.53×10^3
g_3 (circle)	$\pi r^2 = \pi (20)^2$ $= 1256.64$	-80	-40	-100.53×10^3	-50.26×10^3
$\Sigma a = 4456.63$				$\Sigma ay = -313.21 \times 10^3$	$\Sigma ax = -135.61 \times 10^3$

$$\bar{y} = \frac{-313.21 \times 10^3}{4456.63} = \underline{\underline{-70.28 \text{ mm}}}$$

$$\bar{x} = \frac{-135.61 \times 10^3}{4456.63} = \underline{\underline{-30.43 \text{ mm}}}$$

The co-ordinates of the given fig. is $(-30.43 \text{ mm}, -70.28 \text{ mm})$

8. Locate the centroid of the area as shown in fig. with cartesian co-ordinate shown.



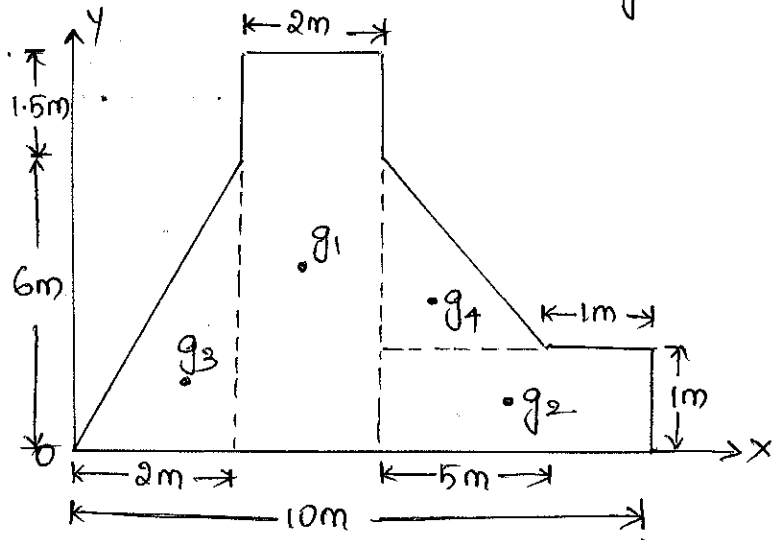
Solⁿ: Consider the reference axis as OX & OY shown in fig.

The fig. is unsymmetrical.

$\therefore$ find $\bar{x}$ & $\bar{y}$

$$\bar{y} = \frac{\sum ay}{\sum a}$$

$$\bar{x} = \frac{\sum ax}{\sum a}$$



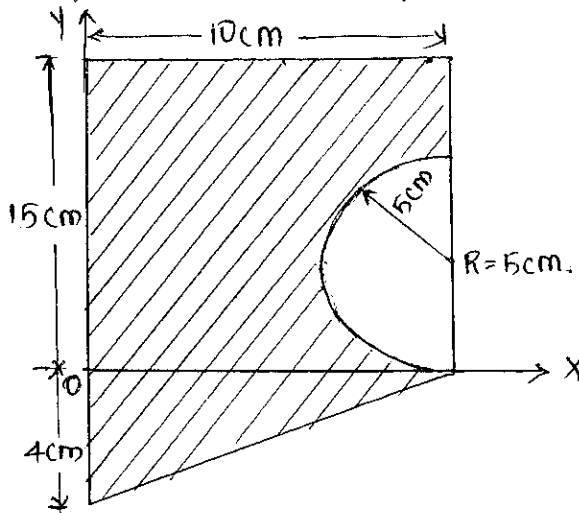
Component	Area (a) m ²	y - m	x - m	ay - m ³	ax - m ³
g_1 (Rectangle)	$2 \times 1.5 = 3$	$\frac{1.5}{2} = 0.75$	$2 + \frac{2}{2} = 3$	$3 \times 0.75 = 2.25$	$3 \times 3 = 9$
g_2 (Rectangle)	$5 \times 1 = 5$	$\frac{1}{2} = 0.5$	$2 + \frac{5}{2} = 4.5$	$5 \times 0.5 = 2.5$	$5 \times 4.5 = 22.5$
g_3 (triangle)	$\frac{1}{2} \times 2 \times 6 = 6$	$\frac{1}{3}(6) = 2$	$\frac{2}{3}(2) = 1.33$	$6 \times 2 = 12$	$6 \times 1.33 = 8$
g_4 (triangle)	$\frac{1}{2} \times 5 \times 1 = 2.5$	$1 + \frac{1}{3}(1) = 1.33$	$2 + \frac{1}{3}(5) = 2.67$	$2.5 \times 1.33 = 3.33$	$2.5 \times 2.67 = 6.68$
	$\sum a = 16.5$			$\sum ay = 104.63$	$\sum ax = 165.88$

$$\bar{y} = \frac{104.63}{16.5} = 6.34$$

$$\bar{x} = \frac{165.88}{16.5} = 10.05$$

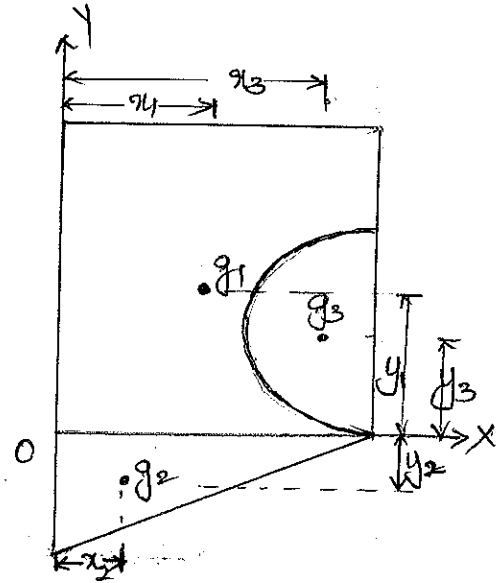
The co-ordinates of the given fig is (10.05m, 6.34m)

9. Locate the centroid of the shaded area shown in fig.



Solⁿ: The given fig is unsymmetrical about both the axis.

$$\therefore \text{find } \bar{y} = \frac{\sum ay}{\sum a} \quad \& \quad \bar{x} = \frac{\sum ax}{\sum a}$$



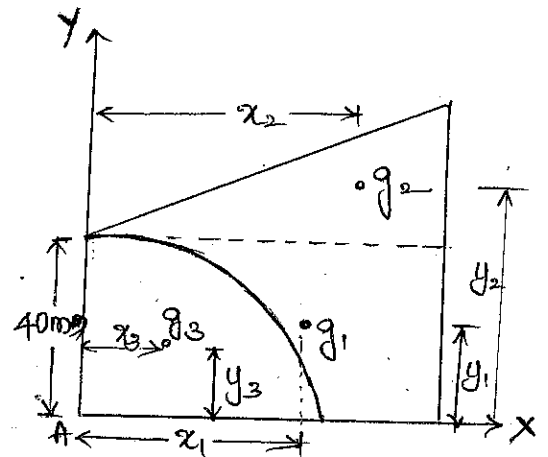
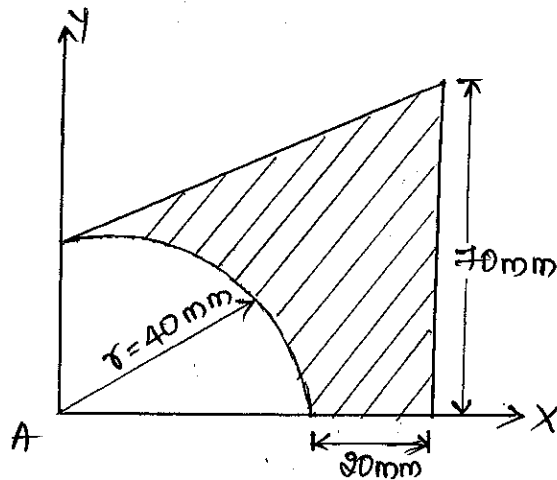
Component	Area(a) cm ²	y - cm	x - cm	ay (cm ³)	ax (cm ³)
g ₁ (Rectangle)	b × d 10 × 15 = 150	$\frac{15}{2} = 7.5$	$\frac{10}{2} = 5$	1125	750
g ₂ (Triangle)	$\frac{1}{2}bh$ $= \frac{1}{2} \times 10 \times 4$ = 20	$-\frac{h}{3} = -\frac{4}{3}$ = -1.33	$\frac{1}{3}(10) = 3.33$	-26.6	66.6
g ₃ (Semi-circle)	$\frac{\pi r^2}{2}$ $= \frac{\pi(5)^2}{2}$ = 39.26	5	$\frac{10 - \frac{48}{3\pi}}{3\pi}$ = 7.87	196.3	308.97
	$\sum a = 130.74$			$\sum ay = 902.1$	$\sum ax = 507.63$

$$\bar{y} = \frac{902.1}{130.74} = 6.9 \text{ cm}$$

$$\bar{x} = \frac{507.63}{130.74} = 3.88 \text{ cm}$$

The co-ordinates of the given fig is (3.88 cm, 6.9 cm)

10. Locate the centroid of the lamina shown in fig. wrt point A. 25



Soln: The fig. is unsymmetrical about x & y axis.

$\therefore$ find $\bar{y} = \frac{\sum ay}{\sum a}$ & $\bar{x} = \frac{\sum ax}{\sum a}$

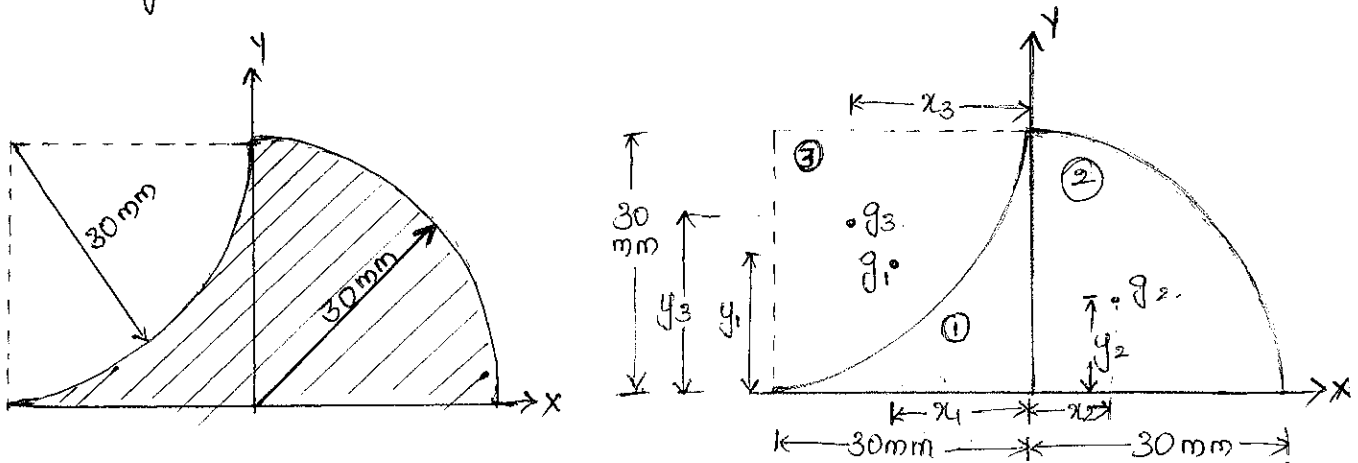
Component	Area(a) mm ²	y in mm	x in mm	ay (mm ³)	ax (mm ³)
g ₁ (Rectangle)	bxd 60x40 = 2400	$\frac{40}{2} = 20$	$\frac{60}{2} = 30$	48×10^3	72×10^3
g ₂ (Triangle)	$\frac{1}{2}bh$ $\frac{1}{2} \times 60 \times 40$ = 1200	$40 + \frac{30}{3}$ = 50	$\frac{2}{3}(60)$ = 40	48×10^3	36×10^3
Deductions					
g ₃ (Quarter circle)	$\frac{\pi r^2}{4} = \frac{\pi (40)^2}{4}$ = 1256.64	$\frac{48}{3\pi}$ = 16.97	$\frac{48}{3\pi}$ = 16.97	21.32×10^3	21.32×10^3
	$\sum a = 2043.36$			$\sum ay = 71.68 \times 10^3$	$\sum ax = 86.68 \times 10^3$

$$\bar{y} = \frac{71.68 \times 10^3}{2043.36} = \underline{\underline{35.08 \text{ mm}}}$$

$$\bar{x} = \frac{86.68 \times 10^3}{2043.36} = \underline{\underline{42.42 \text{ mm}}}$$

The co-ordinates of the given fig is $(42.42 \text{ mm}, 35.08 \text{ mm})$

11. Locate the centroid of the shaded portion as shown in fig.



Solⁿ: The given fig. is unsymmetrical about x & y axis.
 $\therefore$ find $\bar{y} = \frac{\sum ay}{\sum a}$ & $\bar{x} = \frac{\sum ax}{\sum a}$.

Component	Area (a) mm ²	y in mm	x in mm	ay (mm ³)	ax (mm ³)
g ₁ (square)	bd 30 × 30 = 900	$\frac{30}{2} = 15$	$-\frac{30}{2} = -15$	13500	-13500
g ₂ (Quarter circle)	$\frac{\pi r^2}{4} = \frac{\pi (30)^2}{4}$ = 706.85	$\frac{4r}{3\pi} = \frac{4(30)}{3\pi}$ = 12.73	$\frac{4r}{3\pi} = \frac{4(30)}{3\pi}$ = 12.73	8998.2	8998.2
Deduction. g ₃ (Quarter circle)	$\frac{\pi r^2}{4} = \frac{\pi (30)^2}{4}$ = 706.85	$(30 - \frac{4r}{3\pi})$ = 17.27	$-(30 - \frac{4r}{3\pi})$ = -17.27	12207.3	-12207.3
	$\sum a = 900$			$\sum ay = 10290.9$	$\sum ax = 7705.5$

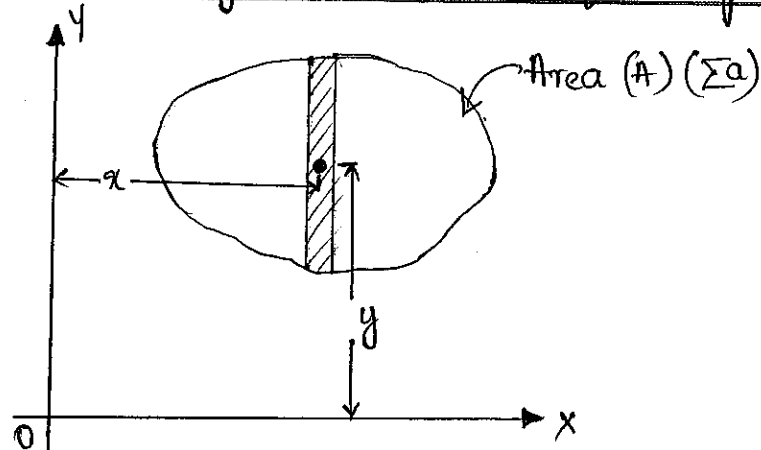
$$\bar{y} = \frac{10290.9}{900} = \underline{\underline{11.43 \text{ mm}}}$$

$$\bar{x} = \frac{7705.5}{900} = \underline{\underline{8.56 \text{ mm}}}$$

The co-ordinates of the given fig. is (8.56 mm, 11.43 mm)

MOMENT OF INERTIA (MI)

Moment of Inertia by the method of Integration.



Consider a plane lamina as shown in fig. whose moment of inertia is to be found out about 'x-x' & 'y-y' axis.

Divide the whole area into no. of strips.

Consider one of the shaded strip.

Let x & y be the distance of C.G. of the strip from y-y & x-x axis.

Now, MI of strip about y-y axis = $x^2 \cdot da$.

MI of strip about x-x axis = $y^2 \cdot da$.

MI of whole lamina may be found by integrating the above eqn.

$$\begin{aligned} I_{yy} &= \int x^2 \cdot da = \Sigma x^2 \cdot da \\ I_{xx} &= \int y^2 \cdot da = \Sigma y^2 \cdot da. \end{aligned}$$

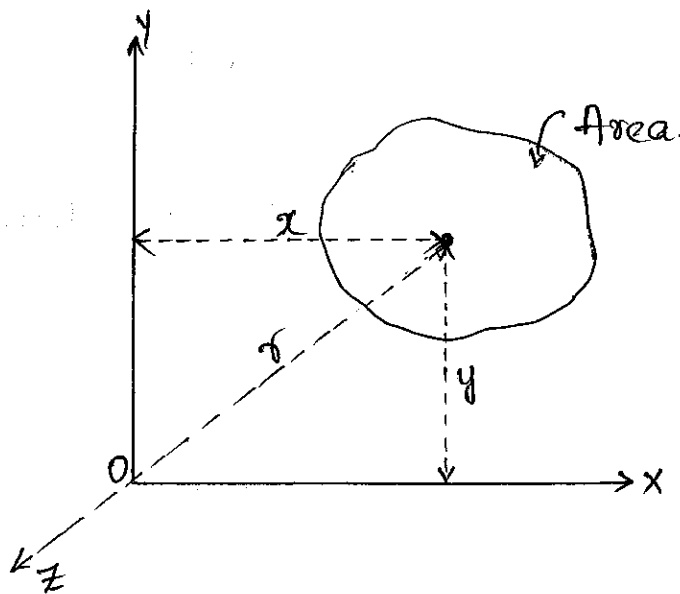
• Moment of Inertia (MI) is expressed in terms of mm^4 .

Definition of Moment of Inertia (MI)

MI is the product of the area (A) & its perpendicular distance (x). The product of Ax is called first moment of area. If this moment of area (Ax) is once again multiplied by the same perpendicular distance (x) i.e., (Ax^2). Then this product is called the moment of Inertia (or) MI.

$$\boxed{I = y^2 \cdot dA} \quad (\text{or}) \quad \boxed{I = x^2 \cdot dA}$$

Polar moment of Inertia (I_{zz} (or) I_p)



If a plane lamina lies in the xy plane as shown in fig. then the axis perpendicular to the xy plane is z -axis. The polar moment of inertia of an area about an axis perpendicular to its plane is known as "Polar moment of Inertia".

Consider an area in xy plane as shown in fig. Then its moment of inertia of an area about perpendicular

axis, z -axis passing through the origin 'O' is given by, $I_{zz} = r^2 \cdot da$

$$I_{zz} = (x^2 + y^2) \cdot da$$

WKT, $r^2 = x^2 + y^2$

$$I_{zz} = x^2 \cdot da + y^2 \cdot da$$

$$I_{zz} = I_{yy} + I_{xx}$$

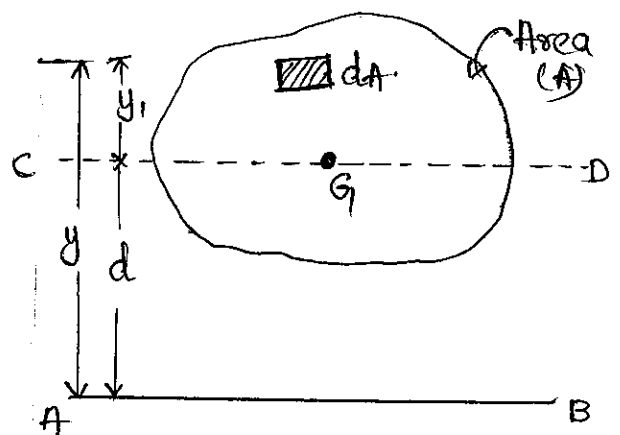
Parallel axis theorem.

Statement :- The moment of inertia (MI) of any area about an axis in its plane is the sum of moment of inertia about a parallel centroidal axis & the product of area & square of the distance between the two parallel axis.

$$I_{AB} = I_G + Ad^2$$

Proof:-

Consider an area 'A' with centroid 'G' as shown in fig. to find MI about AB. Consider an element da at a distance 'y' as shown.



Let CD be the axis parallel to AB passing through the centroid G. Let 'd' be the distance between the two parallel axis AB & CD.

I_{AB} = MI of an area about AB

I_G = MI of an area about CD.

By the definition,

$$I_{AB} = \int y^2 \cdot dA.$$

Here $y = y_1 + d$.

$$\begin{aligned} \therefore I_{AB} &= \int (y_1 + d)^2 \cdot dA \\ &= \int (y_1^2 + d^2 + 2y_1 \cdot d) dA. \end{aligned}$$

$$I_{AB} = \int y_1^2 \cdot dA + \int d^2 \cdot dA + \int 2y_1 \cdot d \cdot dA \longrightarrow (1)$$

The mg of area A about centroidal axis CD is,

$$I_G = \int y_1^2 \cdot dA.$$

Moment of area 'A' about its centroidal axis CD is given

$$\text{by } \int y_1 \cdot dA = 0.$$

$$\text{Also, } \int dA = A.$$

Substituting these values in eqn (1) we get,

$$I_{AB} = I_G + d^2 \cdot A + 0.$$

$$\boxed{I_{AB} = I_G + Ad^2}$$

Hence proof.

==

Perpendicular axis theorem.

Statement : The MI of an area about an axis perpendicular to its plane (polar MI) at any point 'O' is equal to the sum of the moments of inertia about any two mutually perpendicular axis through the same point 'O' & lying in the same plane of area.

$$\boxed{I_{zz} = I_{xx} + I_{yy}}$$

Proof : let us consider an elemental area 'dA' at a distance 'r' from 'O'. Let the co-ordinates of dA be x & y. Then from the definition of polar MI,

$$I_{zz} = \sum r^2 \cdot dA.$$

W.K.T,
 $r^2 = x^2 + y^2$

$$I_{zz} = \sum (x^2 + y^2) \cdot dA.$$

$$I_{zz} = \sum x^2 \cdot dA + \sum y^2 \cdot dA.$$

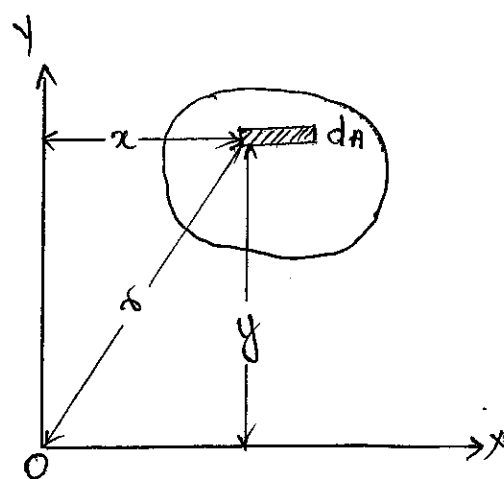
$$I_{zz} = I_{yy} + I_{xx}$$

(or)

$$\boxed{I_{zz} = I_{xx} + I_{yy}}$$

Hence proved.

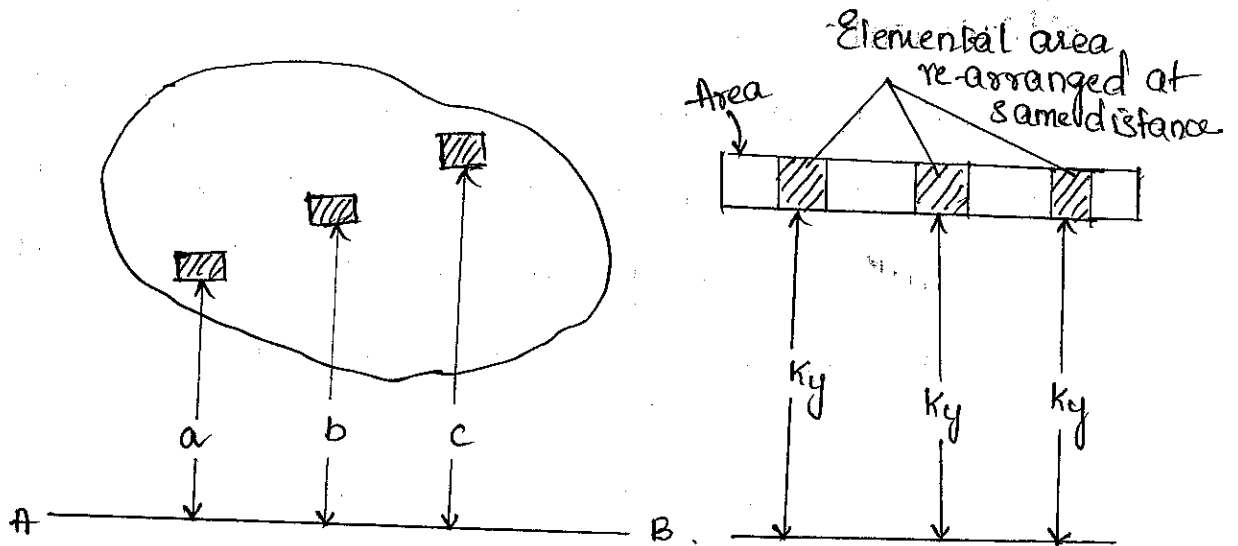
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Here, $\sum y^2 \cdot dA = I_{xx}$

$$\sum x^2 \cdot dA = I_{yy}$$

Radius of Gyration (k)



Radius of gyration of given area about an axis may be defined as the distance from the reference axis where the whole area of the body is assumed to be squeezed & concentrated so as not to alter the MI about the reference axis.

Mathematically,

$$\text{Radius of gyration (k)} = \sqrt{\frac{I}{A}} \quad (\text{or}) \quad I = Ak^2$$

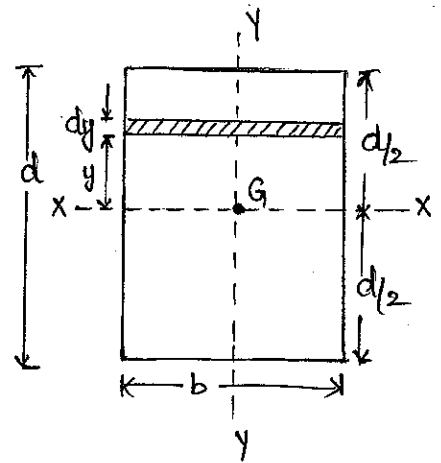
$$(\text{or}) \quad k_{xx} = \sqrt{\frac{I_{xx}}{A}}$$

$$\& \quad k_{yy} = \sqrt{\frac{I_{yy}}{A}}$$

Moment of Inertia by the method of integration.

Moment of Inertia of a Rectangle.

Consider a rectangle of dimensions $b \times d$ as shown in fig. Let 'G' be the centroid of the rectangular area.



Consider an elemental strip 'dA' of width 'b' & length 'dy' at a distance 'y' from the centroidal axis.

By the definition of MI, $I_{xx} = \int y^2 \cdot dA$. $\longrightarrow (1)$

Here, $dA = b \cdot dy$.

On substitution in eqn (1)

$$I_{xx} = \int y^2 \cdot b \cdot dy$$

$$I_{xx} = \int_{-d/2}^{d/2} y^2 \cdot b \cdot dy$$

$$I_{xx} = b \cdot \int_{-d/2}^{d/2} y^2 \cdot dy$$

$$I_{xx} = b \cdot \left[\frac{y^3}{3} \right]_{-d/2}^{d/2}$$

$$I_{xx} = \frac{b}{3} \left[\frac{d^3}{8} + \frac{d^3}{8} \right]$$

$$I_{xx} = \frac{b}{3} \left[\frac{2d^3}{8} \right]$$

$$I_{xx} = \frac{bd^3}{12}$$

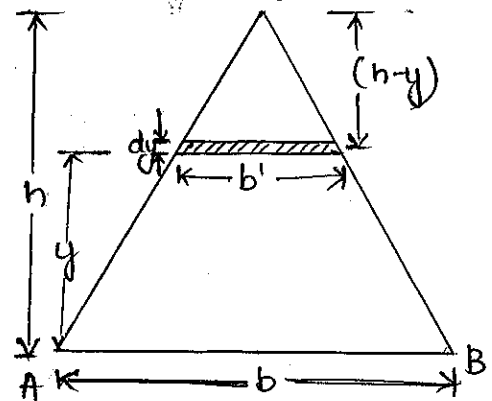
Similarly,

$$I_{yy} = \frac{db^3}{12}$$

Moment of Inertia of Triangle

Consider a triangle of base 'b' & height 'h' as shown in fig.

Choose a horizontal strip of length 'b'' & width 'dy' at a distance 'y' from the base AB.



By similarity of triangles,

$$\frac{b'}{b} = \frac{h-y}{h}.$$

$$b' = \frac{b(h-y)}{h}.$$

$$dA = b' \cdot dy.$$

$$dA = \frac{b(h-y)}{h} \cdot dy$$

MI of whole triangle about AB, = $y^2 \cdot dA$

$$I_{AB} = \int_0^h y^2 \cdot dA.$$

$$I_{AB} = \int_0^h \frac{b(h-y)}{h} \cdot y^2 \cdot dy.$$

$$I_{AB} = \frac{b}{h} \int_0^h hy^2 - y^3 \cdot dy.$$

$$I_{AB} = \frac{b}{h} \left[\frac{h \cdot y^3}{3} - \frac{y^4}{4} \right]_0^h$$

$$I_{AB} = \frac{b}{h} \left[\frac{h^4}{3} - \frac{h^4}{4} \right]$$

$$I_{AB} = \frac{b}{h} \left[\frac{h^4}{12} \right]$$

$$I_{AB} = \frac{bh^3}{12}$$

To find MI about centroid G,

$$I_{AB} = I_G + A\bar{y}^2$$

$$\frac{bh^3}{12} = I_G + \frac{1}{2}bh \left(\frac{h}{3} \right)^2$$

$$\frac{bh^3}{12} = I_G + \frac{bh^3}{18}$$

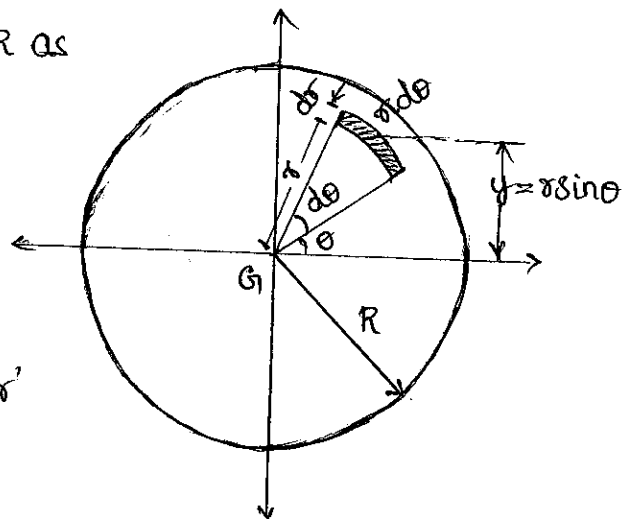
$$I_G = \frac{bh^3}{12} - \frac{bh^3}{18}$$

$$I_G = \frac{bh^3}{36}$$

Here,
 $\bar{y} = \frac{1}{3}(h)$

Moment of Inertia for Circle

Consider a circle of radius R as shown in fig. Choose a differential element of angular width ' $d\theta$ ' at an angle ' θ ' & radial width ' dr ' at a distance ' r ' from the centre as shown in fig.



Area of elemental strip, $dA = (dr)(r.d\theta)$

$$\text{i.e., } dA = r.dr.d\theta.$$

The distance of element from centroidal x -axis,

$$y = r \sin \theta.$$

By the defⁿ of I_x , $I_{xx} = \int_0^R y^2.dA.$

$$I_{xx} = \int_0^R (r^2 \sin^2 \theta) (r.dr.d\theta)$$

$$= \int_0^R r^3 \sin^2 \theta . dr.d\theta$$

$$= \int_0^R \left[\int_0^{2\pi} \sin^2 \theta . d\theta \right] . r^3 . dr.$$

$$= \int_0^R \left[\int_0^{2\pi} \left(\frac{1 - \cos 2\theta}{2} \right) . d\theta \right] . r^3 . dr.$$

$$= \int_0^R \frac{1}{2} \left[\theta - \frac{\sin 2\theta}{2} \right]_0^{2\pi} . r^3 . dr.$$

$$= \frac{1}{2} \int_0^R \left[2\pi - \frac{\sin 2(2\pi)}{2} \right] r^3 . dr.$$

$$= \frac{1}{2} \times 2\pi \left[\frac{r^4}{4} \right]_0^R.$$

$$\boxed{I_G \text{ (or) } I_{xx} = \frac{\pi R^4}{4}}$$

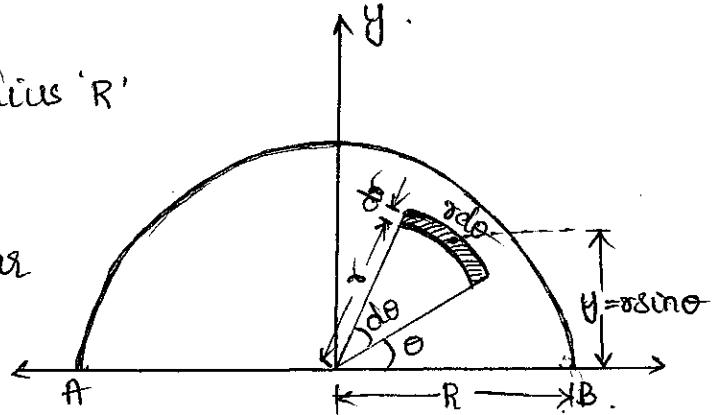
$$\text{or } I_G \text{ or } I_{yy} = \frac{\pi R^4}{4}$$

But $R = D/2$.

$$I_{xx} = I_{yy} = \frac{\pi D^4}{64}$$

Moment of Inertia for Semicircle.

Consider a semicircle of radius 'R' as shown in fig. Choose differential element of angular width ' $d\theta$ ' at an angle ' θ ' & radial width ' dr ' at a distance ' r ' from the centre.



Area of the differential element, $dA = (r \cdot d\theta) dr$.

i.e., $dA = r \cdot dr \cdot d\theta$.

The distance of the differential element from the x-axis is, $y = r \sin \theta$.

$\therefore$ Moment of Inertia, $I_{AB} = \int y^2 \cdot dA$

$$I_{AB} = \int_0^R \int_0^\pi (r \sin \theta)^2 \cdot (r \cdot dr \cdot d\theta)$$

$$I_{AB} = \int_0^R r^3 \cdot \sin^2 \theta \cdot d\theta \cdot dr$$

$$I_{AB} = \int_0^R \left[\int_0^\pi \sin^2 \theta \cdot d\theta \right] r^3 \cdot dr$$

$$I_{AB} = \int_0^R \left[\int_0^\pi \frac{1 - \cos 2\theta}{2} d\theta \right] r^3 dr$$

$$= \frac{1}{2} \int_0^R \left[\theta - \frac{\sin 2\theta}{2} \right]_0^\pi r^3 dr$$

$$= \frac{1}{2} \int_0^R (\pi - 0) r^3 dr$$

$$= \frac{1}{2} \cdot \pi \cdot \frac{R^4}{4}$$

$$\boxed{I_{AB} = \frac{\pi R^4}{8}}$$

Centroid is at a distance of $\frac{4R}{3\pi}$ from AB.

WKT, $I_{AB} = I_G + A\bar{y}^2$

$$\frac{\pi R^4}{8} = I_G + \left(\frac{\pi R^2}{2} \right) \left(\frac{4R}{3\pi} \right)^2$$

$$\left\{ \begin{array}{l} A = \frac{\pi R^2}{2} \\ \bar{y} = \frac{4R}{3\pi} \end{array} \right.$$

$$\frac{\pi R^4}{8} = I_G + \frac{16\pi R^4}{18\pi^2}$$

$$I_G = \frac{\pi R^4}{8} - \frac{16\pi R^4}{18\pi^2}$$

$$I_G = R^4 \left[\frac{\pi}{8} - \frac{8}{9\pi} \right]$$

$$\boxed{I_G = 0.11R^4}$$

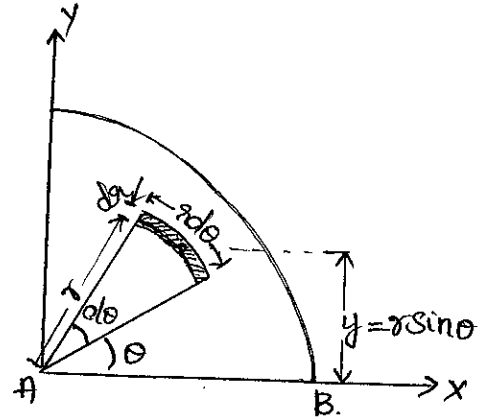
If, $R = D/2$.

$$\boxed{I_G = 0.00686D^4}$$

Moment of Inertia for quarter circle.

Consider a quarter circle of radius 'R' as shown in fig.

Choose a differential element of angular width ' $d\theta$ ' at an angle ' θ ' from AB & radial width ' dr ' at a distance ' r ' from centre.



The area of differential element will be,

$$dA = (r \cdot d\theta) \cdot dr$$

$$dA = r \cdot dr \cdot d\theta$$

The distance of this differential element from AB is,

$$y = r \sin \theta$$

Moment of inertia is, $I_{AB} = \int y^2 \cdot dA$

$$I_{AB} = \int_0^R \int_0^{\pi/2} (r \sin \theta)^2 \cdot (r \cdot dr \cdot d\theta)$$

$$I_{AB} = \int_0^R \int_0^{\pi/2} r^3 \sin^2 \theta \cdot dr \cdot d\theta$$

$$= \int_0^R \left[\int_0^{\pi/2} \frac{1 - \cos 2\theta}{2} \cdot d\theta \right] \cdot r^3 \cdot dr$$

$$= \frac{1}{2} \int_0^R \left[\theta - \frac{\sin 2\theta}{2} \right]_0^{\pi/2} \cdot r^3 \cdot dr$$

$$= \frac{1}{2} \int_0^R \frac{\pi}{2} \cdot r^3 \cdot dr$$

$$I_{AB} = \frac{\pi}{4} \left(\frac{d^4}{4} \right)_0^R$$

$$\boxed{I_{AB} = \frac{\pi R^4}{16}}$$

WKT, $I_{AB} = I_G + A\bar{y}^2$

$$\frac{\pi R^4}{16} = I_G + \left(\frac{\pi R^2}{4} \right) \left(\frac{4R}{3\pi} \right)^2 \quad A = \frac{\pi R^2}{4} ; \bar{y} = \frac{4R}{3\pi}$$

$$\frac{\pi R^4}{16} = I_G + \frac{16\pi R^4}{36\pi^2}$$

$$I_G = \frac{\pi R^4}{16} - \frac{4R^4}{9\pi}$$

$$I_G = R^4 \left[\frac{\pi}{16} - \frac{4}{9\pi} \right]$$

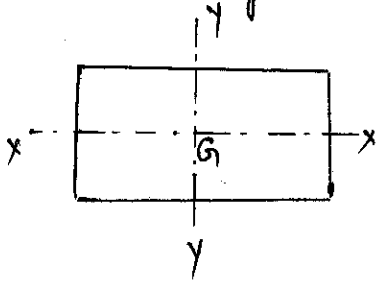
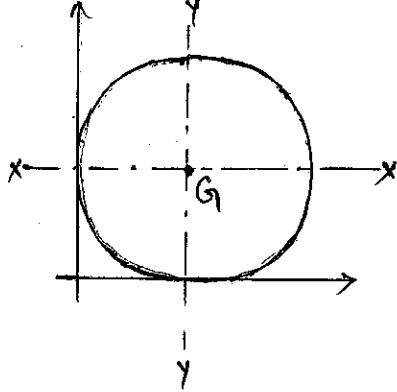
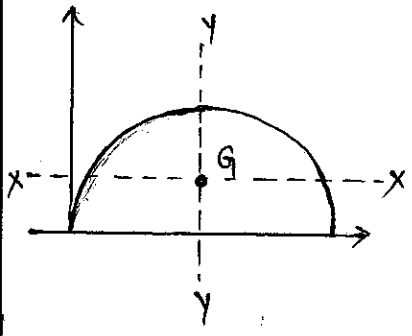
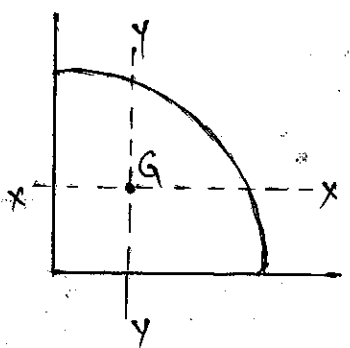
$$\boxed{I_G = 0.055R^4}$$

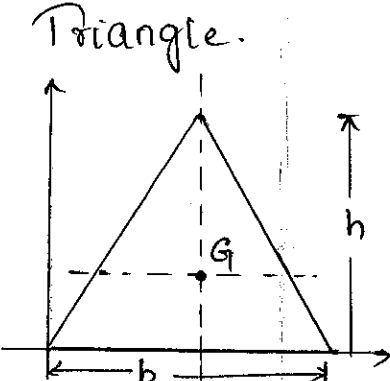
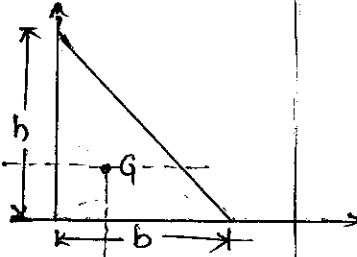
If $R = D/2$.

(or)

$$\boxed{I_G = 0.00348D^4}$$

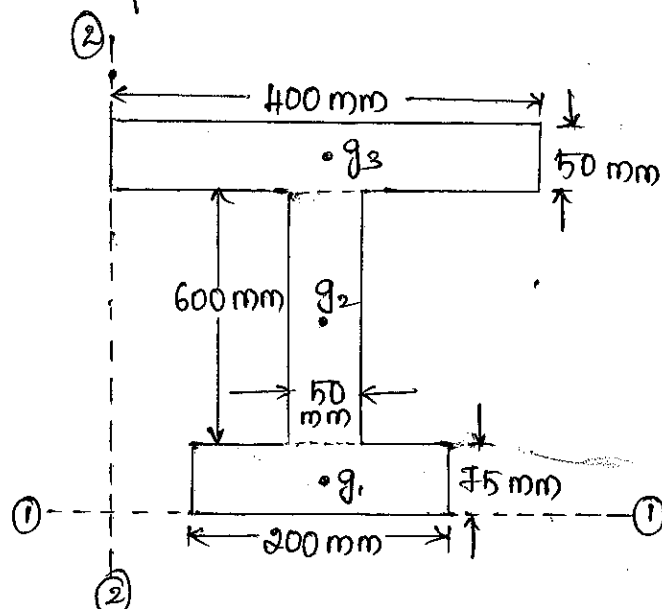
Table showing MP of some common geometrical figures.

Sl.no.	Shape.	I_{xx}	I_{yy}
1.	Rectangle 	$\frac{bd^3}{12}$	$\frac{db^3}{12}$
2.	Circle (radius = r) 	$\frac{\pi r^4}{4}$ (or) $\frac{\pi D^4}{64}$	$\frac{\pi r^4}{4}$ (or) $\frac{\pi D^4}{64}$
3.	Semicircle (radius = r) 	0.1104	$\frac{\pi r^4}{8}$
4.	Quarter circle (radius = r) 	0.05584	0.05584

Sl. No.	Shape	I_{xx}	I_{yy}
5.	Triangle. 	$\frac{bh^3}{36}$	$\frac{hb^3}{48}$
6.	Right angled triangle 	$\frac{bh^3}{36}$	$\frac{hb^3}{36}$

Problems.

1. Find the I_x along the horizontal axis passing through the centroid of the section shown in fig.



Solⁿ: Consider the reference axis 1-1 & 2-2 as shown in fig. The fig. is symmetrical about y -axis. $\therefore$ It is required to find I_x .

$$\bar{y} = \frac{\sum ay}{\sum a}$$

$$\bar{x} = \frac{400}{2} = 200 \text{ mm.}$$

Component	Area (mm ²)	Distance of 'G' from (1)-(1), y	ay (mm ³)	ay ² (mm ⁴)	I _{gx} (mm ⁴)
g ₁ (rectangle)	bxd 200x75 = 15x10 ³	$\frac{75}{2} = 37.5$	562.5x10 ³	21.09x10 ⁶	$\frac{bd^3}{12}$ = $\frac{200 \times 75^3}{12}$ = 7.03x10 ⁶
g ₂ (rectangle)	bxd 50x600 = 30x10 ³	$75 + \frac{600}{2}$ = 375	11.25x10 ⁶	4.21x10 ⁹	$\frac{bd^3}{12}$ = $\frac{50 \times 600^3}{12}$ = 900x10 ⁶
g ₃ (rectangle)	bxd 400x50 = 20x10 ³	$75 + 600 + \frac{50}{2}$ = 700	14x10 ⁶	9.8x10 ⁹	$\frac{bd^3}{12}$ = $\frac{400 \times 50^3}{12}$ = 4.16x10 ⁶
	$\sum a =$ 65x10 ³		$\sum ay =$ 25.81x10 ⁶	$\sum ay^2 =$ 1.403x10 ¹⁰	$\sum I_{gx} =$ 911.19x10 ⁶

$$\bar{y} = \frac{25.81 \times 10^6}{65 \times 10^3} = \underline{\underline{397.1 \text{ mm.}}}$$

The co-ordinates of centroid are (400mm, 397.1 mm)

$$\begin{aligned} I_{1-1} &= \sum I_{gx} + \sum ay^2 \\ &= (911.19 \times 10^6) + (1.403 \times 10^{10}) \\ &= \underline{\underline{1.49 \times 10^{10} \text{ mm}^4}} \end{aligned}$$

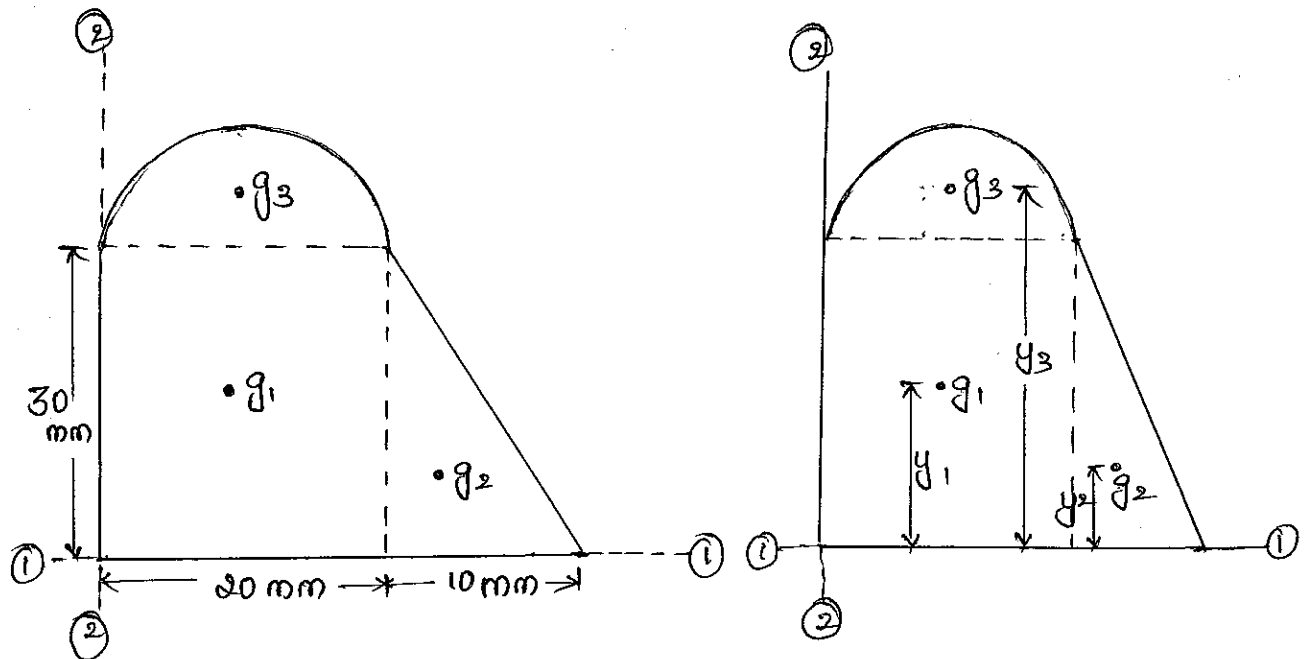
W.K.P, $I_{1-1} = I_{xx} + A\bar{y}^2$

i.e., $I_{xx} = I_{1-1} - A\bar{y}^2$

$$I_{xx} = (1.49 \times 10^{10}) - (65 \times 10^3)(397.1)^2$$

$$I_{xx} = 4.65 \times 10^9 \text{ mm}^4$$

2. Find the ME of the fig. shown, about the horizontal axis.



Solⁿ: Consider the reference axis as shown in fig.
It is required to find $\bar{y} = \frac{\sum ay}{\sum a}$.

Component	Area (mm ²)	y in mm	ay (mm ³)	ay ² (mm ⁴)	I _{gx} (mm ⁴)
g ₁ (rectangle)	20 × 30 = 600	$\frac{30}{2} = 15$	9000	135 × 10 ³	$\frac{bd^3}{12}$ = 45 × 10 ³
g ₂ (triangle)	$\frac{1}{2} \times 10 \times 30$ = 150	$\frac{30}{3} = 10$	1500	15 × 10 ³	$\frac{bh^3}{36}$ = $\frac{10 \times 30^3}{36}$ = 7500
g ₃ (semicircle)	$\frac{\pi r^2}{2} = \frac{\pi (10)^2}{2}$ = 157	$30 + \frac{48}{3\pi}$ = 34.24	5375.7	184.06 × 10 ³	0.1184 = 0.11(10) ⁴ = 1100
	Σa = 907		Σay = 15875.7	Σay ² = 334.06 × 10 ³	ΣI _{gx} = 53.6 × 10 ³

$$\bar{y} = \frac{15875.7}{907} = \underline{\underline{17.5 \text{ mm}}}$$

$$\begin{aligned} I_{1-1} &= \sum I_{gx} + \sum ay^2 \\ &= (53.6 \times 10^3) + (334.06 \times 10^3) \\ &= \underline{\underline{387.66 \times 10^3 \text{ mm}^4}} \end{aligned}$$

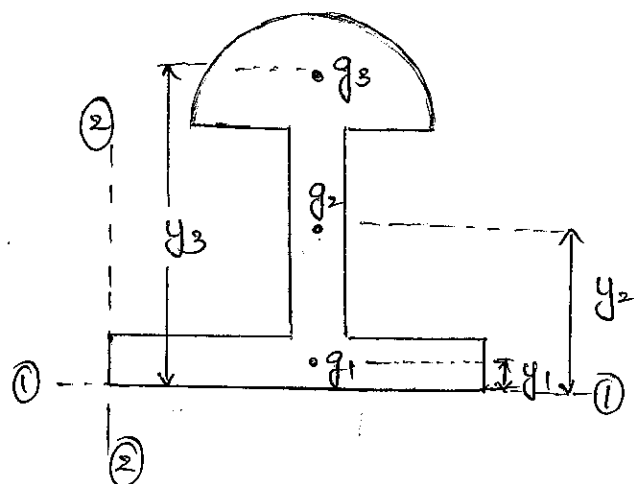
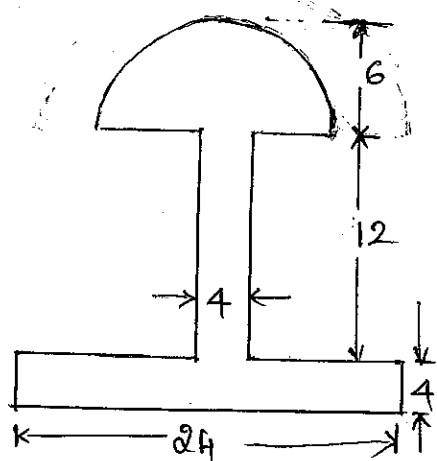
$$\text{W.K.P, } I_{1-1} = I_{xx} + A\bar{y}^2$$

$$I_{xx} = I_{1-1} - A\bar{y}^2$$

$$= (387.66 \times 10^3) - (907 \times 17.5^2)$$

$$\boxed{I_{xx} = 109.89 \times 10^3 \text{ mm}^4}$$

3. Calculate the M.I. along the horizontal axis passing through the centroid of the section as shown in fig. Also find the radius of gyration. All dimensions are in mm.



Solⁿ: Consider the reference axes ①-① & ②-② as shown in fig.

$$\text{To find } \bar{y} = \frac{\sum ay}{\sum a}$$

$$\text{To find radius of gyration, } k = \sqrt{\frac{I_p}{A}}$$

where I_p = Polar M.I. given by $I_p = I_{zz} = I_{xx} + I_{yy}$

Component	Area (mm ²)	y - (mm)	ay (mm ³)	ay ² (mm ⁴)	I _{gx} (mm ⁴)
g ₁ (rectangle)	bd 24 × 4 = 96	$\frac{4}{2} = 2$	192	384	$\frac{bd^3}{12} = \frac{24 \times 4^3}{12}$ = 128
g ₂ (rectangle)	4 × 12 = 48	$4 + \frac{12}{2} = 10$	480	4800	$\frac{4 \times 12^3}{12}$ = 576
g ₃ (Semicircle)	$\frac{\pi r^2}{2}$ $\frac{\pi (6)^2}{2}$ = 56.55	$4 + 12 + \frac{48}{3\pi}$ = 18.55	1049	19.45×10^3	0.1184 = 142.56
	$\Sigma a = 200.55$		$\Sigma ay = 1721$	$\Sigma ay^2 = 24.63 \times 10^3$	$\Sigma I_{gx} = 846.56$

$$\bar{Y} = \frac{1721}{200.55} = 8.58 \text{ mm.}$$

$$\begin{aligned} I_{1-1} &= \Sigma I_{gx} + \Sigma ay^2 \\ &= 846.56 + (24.63 \times 10^3) \\ &= 25.46 \times 10^3 \text{ mm}^4. \end{aligned}$$

$$\begin{aligned} I_{xx} &= I_{1-1} - A\bar{Y}^2 \\ &= (25.46 \times 10^3) - (200.55 \times 8.58^2) \end{aligned}$$

$$I_{xx} = 10.69 \times 10^3 \text{ mm}^4$$

$$\begin{aligned} I_{yy} &= I_{g1y} + I_{g2y} + I_{g3y} \\ &= \frac{db^3}{12} + \frac{db^3}{12} + \frac{\pi r^4}{8} \end{aligned}$$

$$= \frac{4 \times 24^3}{12} + \frac{12 \times 4^3}{12} + \frac{\pi (6)^4}{8}$$

$$\Rightarrow I_{yy} = 5180.94 \text{ mm}^4$$

Polar MP , $I_p = I_{zz} = I_{xx} + I_{yy}$

$$= (10.49 \times 10^3) + (5180.94)$$

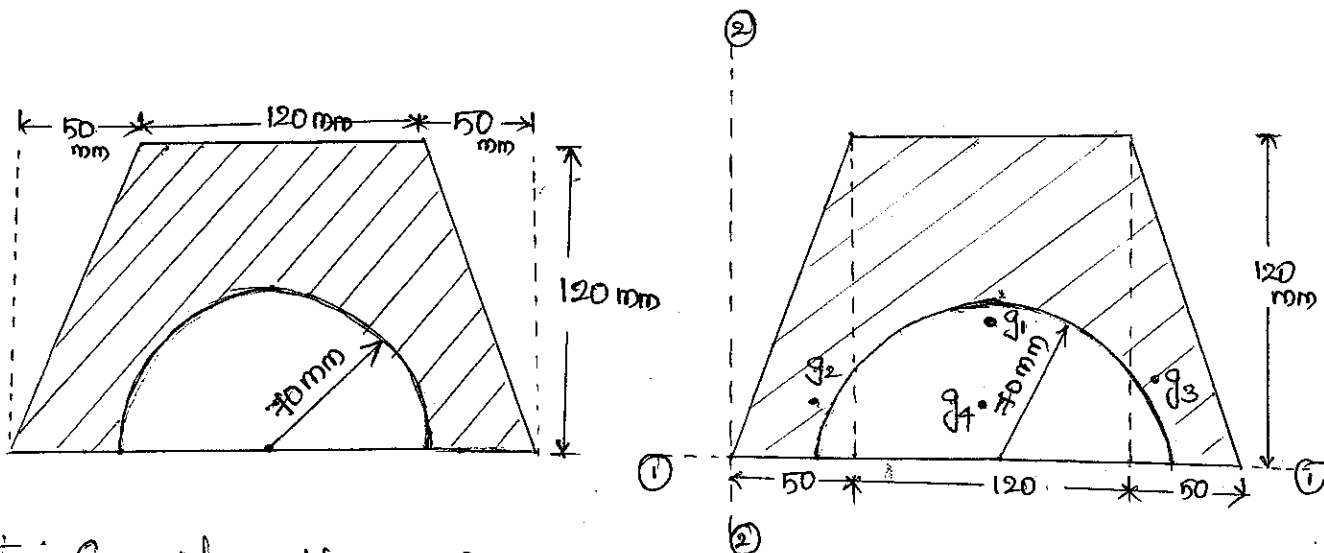
$$I_{zz} = 15.67 \times 10^3 \text{ mm}^4$$

Polar radius of gyration, $K_p = \sqrt{\frac{I_p}{A}}$

$$= \sqrt{\frac{15.67 \times 10^3}{200.55}}$$

$$K_p = 8.84 \text{ mm}$$

4. Determine the MP about the horizontal axis q
also find the radius of gyration about the horizontal axis.



Soln: Consider the reference axis (1)-(1) & (2)-(2) as shown in fig.

To find $\bar{y} = \frac{\sum ay}{\sum a}$

Also, $I_{1-1} = \sum ay^2 + \sum I_{gx}$

$$I_{xx} = I_{1-1} - A\bar{y}^2$$

$$K_{xx} = \sqrt{\frac{I_{xx}}{A}}$$

Component	Area (mm ²)	y - mm	ay - mm ³	ay ² - mm ⁴	I _{gx} - mm ⁴
g ₁ (rectangle)	bd 120 × 120 = 14400	$\frac{120}{2} = 60$	8640 × 10 ³	51.84 × 10 ⁶	$\frac{bd^3}{12}$ = $\frac{120 \times 120^3}{12}$ = 17.28 × 10 ⁶
g ₂ (triangle)	$\frac{1}{2}bh$ $\frac{1}{2} \times 50 \times 120$ = 3000	$\frac{120}{3} = 40$	120 × 10 ³	4.8 × 10 ⁶	$\frac{bh^3}{36} = \frac{50 \times 120^3}{36}$ = 2.4 × 10 ⁶
g ₃ (triangle)	3000	40	120 × 10 ³	4.8 × 10 ⁶	= 2.4 × 10 ⁶
Deduction g ₄ (semicircle)	$\frac{\pi r^2}{2} = \frac{\pi \times 70^2}{2}$ = 7697	$\frac{4r}{3\pi} = \frac{4 \times 70}{3\pi}$ = 29.7	228.67 × 10 ³	6.79 × 10 ⁶	0.1184 = 0.11(70) ⁴ = 2.64 × 10 ⁶
	$\Sigma a =$ 12703		$\Sigma ay =$ 875.33 × 10 ³	$\Sigma ay^2 =$ 54.65 × 10 ⁶	$\Sigma I_{gx} =$ 19.44 × 10 ⁶

$$\bar{y} = \frac{875.33 \times 10^3}{12703} = \underline{\underline{68.9 \text{ mm}}}$$

$$I_{1-1} = (54.65 \times 10^6) + (19.44 \times 10^6)$$

$$= \underline{\underline{74.09 \times 10^6 \text{ mm}^4}}$$

$$I_{xx} = (74.09 \times 10^6) - (12703)(68.9)^2$$

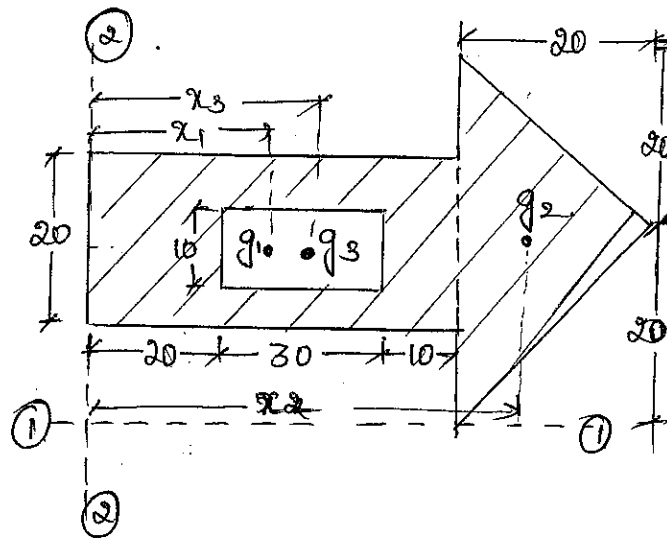
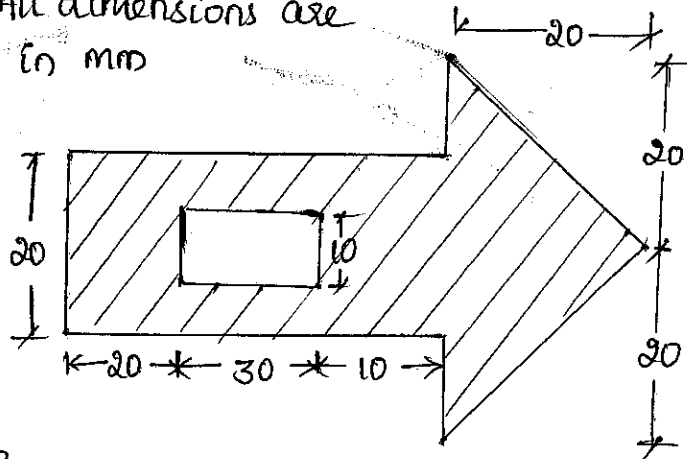
$$I_{xx} = 13.78 \times 10^6 \text{ mm}^4$$

$$K_{xx} = \sqrt{\frac{13.78 \times 10^6}{12703}}$$

$$K_{xx} = \underline{\underline{32.94 \text{ mm}}}$$

5. Find the radius of gyration of the shaded area about an axis normal to symmetrical axis.

All dimensions are in mm



Solⁿ: Consider the reference

axis ①-① & ②-② as shown in fig. The fig. is symmetrical about x-axis. $\therefore$ It is required to find $\bar{x} = \frac{\sum ax}{\sum a}$, $\bar{y} = 20$ mm

Required to find I_x about the axis $\perp$ to x-x (symmetrical) i.e., to find I_{gy} .

Component	Area (mm ²)	x (mm)	ax (mm ³)	ax^2 (mm ⁴)	I_{gy} (mm ⁴)
g_1 (rectangle)	60×20 $= 1200$	$\frac{60}{2} = 30$	36×10^3	1.08×10^6	$\frac{db^3}{12} = \frac{20 \times 60^3}{12}$ $= 360 \times 10^3$
g_2 (Triangle)	$\frac{1}{2} \times 40 \times 20$ $= 400$	$60 + \frac{20}{3}$ $= 66.67$	26.66×10^3	1.77×10^6	$\frac{bh^3}{36} = \frac{40 \times 20^3}{36}$ $= 8.88 \times 10^3$
<u>Deduction</u> g_3 (rectangle)	30×10 $= 300$	$20 + \frac{30}{2}$ $= 35$	10.5×10^3	367.5×10^3	$\frac{db^3}{12} = \frac{10 \times 30^3}{12}$ $= 22.5 \times 10^3$
	$\sum a = 1300$		$\sum ax = 52.16 \times 10^3$	$\sum ax^2 = 2.48 \times 10^6$	$\sum I_{gy} = 346.38 \times 10^3$

$$\bar{x} = \frac{52.16 \times 10^3}{1300} = \underline{\underline{40.12 \text{ mm}}}$$

$$\begin{aligned} I_{2-2} &= \sum ax^2 + \sum I_{gy} \\ &= (2.48 \times 10^6) + (346.38 \times 10^3) \end{aligned}$$

$$\boxed{I_{2-2} = 2.82 \times 10^6 \text{ mm}^4}$$

$$\begin{aligned} I_{yy} &= I_{2-2} - A\bar{x}^2 \\ &= (2.82 \times 10^6) - (1300)(40.12)^2 \end{aligned}$$

$$\boxed{I_{yy} = 727.5 \times 10^3 \text{ mm}^4}$$

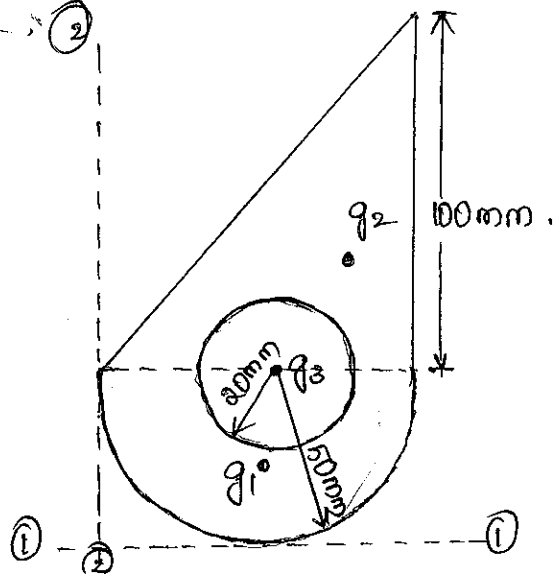
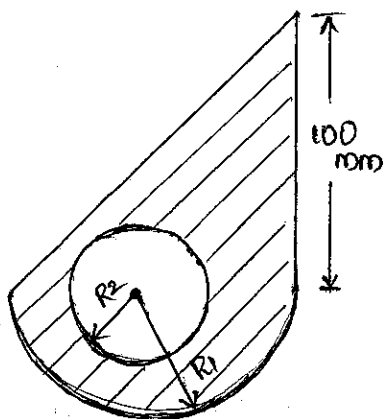
Radius of gyration about y-y axis,

$$k_{yy} = \sqrt{\frac{I_{yy}}{A}} = \sqrt{\frac{727.5 \times 10^3}{1300}}$$

$$\boxed{k_{yy} = 23.65 \text{ mm}}$$

6. Determine the second moment of inertia about the horizontal centroidal axis for the shaded area shown. Also find the radius of gyration about the same axis.

Take $R_1 = 50 \text{ mm}$ & $R_2 = 20 \text{ mm}$; ②



Soln:- Consider the reference axis (1)-(1) & (2)-(2) as shown in fig. It is required to find, $\bar{y} = \frac{\sum ay}{\sum a}$.

$$I_{1-1} = \sum ay^2 + \sum I_{gx}$$

$$I_{x-x} = I_{1-1} - A\bar{y}^2$$

$$K_{xx} = \sqrt{\frac{I_{xx}}{A}}$$

Component	Area (mm ²)	y - mm	ay (mm ³)	ay ² (mm ⁴)	I _{gx} (mm ⁴)
g ₁ (Semicircle)	$\frac{\pi r^2}{2}$ $\frac{\pi (50)^2}{2}$ = 3927	$50 - \frac{4r}{3\pi}$ = 28.8	113.09×10^3	3.25×10^6	0.1184 $= 0.11 (50)^4$ $= 687.5 \times 10^3$
g ₂ (Triangle)	$\frac{1}{2} \times 100 \times 100$ = 5000	$50 + \frac{100}{3}$ = 83.33	416.65×10^3	34.72×10^6	$\frac{bh^3}{36}$ $\frac{100 \times 100^3}{36}$ $= 2.77 \times 10^6$
<u>Deduction</u> g ₃ (circle)	$\pi r^2 = \pi (20)^2$ = 1256.6	50	62.83×10^3	3.14×10^6	$\frac{\pi r^4}{4} = \frac{\pi (20)^4}{4}$ = 125.66×10^3
	$\sum a =$ 7670.4		$\sum ay =$ 466.9×10^3	$\sum ay^2 =$ 34.83×10^6	$\sum I_{gx} =$ 3.33×10^6

$$\bar{y} = \frac{466.9 \times 10^3}{7670.4} = \underline{\underline{60.87 \text{ mm}}}$$

$$I_{1-1} = (34.83 \times 10^6) + (3.33 \times 10^6) = \underline{\underline{38.16 \times 10^6 \text{ mm}^4}}$$

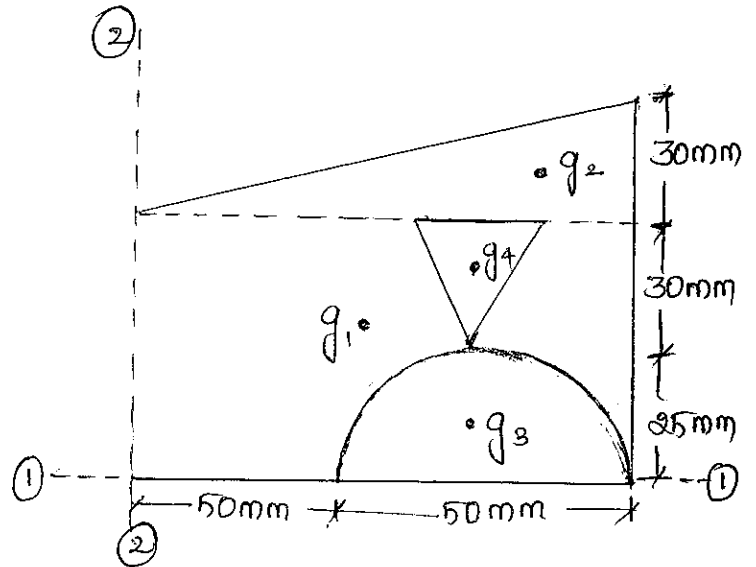
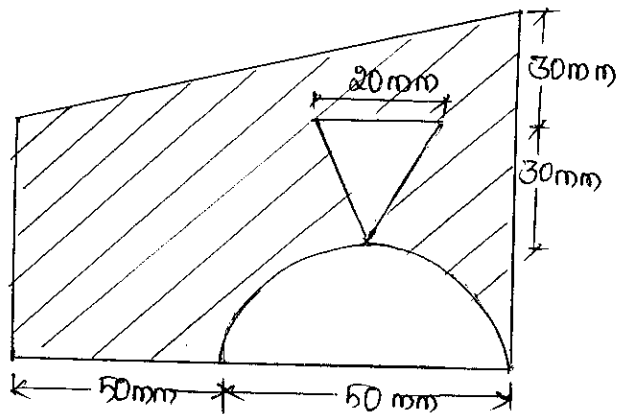
$$I_{xx} = (38.16 \times 10^6) - (7670.4)(60.87)^2$$

$$\boxed{I_{xx} = 9.73 \times 10^6 \text{ mm}^4}$$

$$K_{xx} = \sqrt{\frac{I_{xx}}{A}} = \sqrt{\frac{9.73 \times 10^6}{7670.4}}$$

$$\Rightarrow \boxed{K_{xx} = 35.63 \text{ mm}}$$

7. Find the radius of gyration about x-x axis of the shaded area shown in fig.



Solⁿ: Consider the reference axis (1)-(1) & (2)-(2) as shown in fig.

$$\bar{y} = \frac{\sum ay}{\sum a}$$

$$I_{xx} = I_{1-1} - A\bar{y}^2$$

$$K_{xx} = \sqrt{\frac{I_{xx}}{A}}$$

$$I_{1-1} = \sum ay^2 + \sum I_{gx}$$

Component	Area (mm ²)	y - mm	ay - mm ³	ay ² - mm ⁴	I _{gx} - mm ⁴
g ₁ (rectangle)	100 × 55 = 5500	$\frac{55}{2} = 27.5$	151.25 × 10 ³	4.15 × 10 ⁶	$\frac{bd^3}{12} = \frac{100 \times 55^3}{12}$ = 1.38 × 10 ⁶
g ₂ (triangle)	$\frac{1}{2} \times 100 \times 30$ = 1500	55 + 30/3 = 65	97.5 × 10 ³	6.33 × 10 ⁶	$\frac{bh^3}{36} = \frac{100 \times 30^3}{36}$ = 45 × 10 ³
g ₃ (Semicircle)	$\frac{\pi r^2}{2} = \frac{\pi (25)^2}{2}$ = 981.8	$\frac{4r}{3\pi} = \frac{4 \times 25}{3\pi}$ = 10.6	10.4 × 10 ³	110.31 × 10 ³	0.1184 0.11(25) ⁴ = 42.96 × 10 ³
g ₄ (triangle)	$\frac{1}{2} \times 20 \times 30$ = 300	25 + $\frac{2}{3}(30)$ = 45	13.5 × 10 ³	607.5 × 10 ³	$\frac{20 \times 30^3}{36}$ = 15 × 10 ³
$\sum a =$	5718.2		$\sum ay =$ 224.85 × 10 ³	$\sum ay^2 =$ 9.76 × 10 ⁶	$\sum I_{gx} =$ 1.39 × 10 ⁶

$$\bar{y} = \frac{224.85 \times 10^3}{5718.2} = \underline{\underline{39.32 \text{ mm}}}$$

$$\begin{aligned} I_{1-1} &= (9.76 \times 10^6) + (1.39 \times 10^6) \\ &= \underline{\underline{11.15 \times 10^6 \text{ mm}^4}} \end{aligned}$$

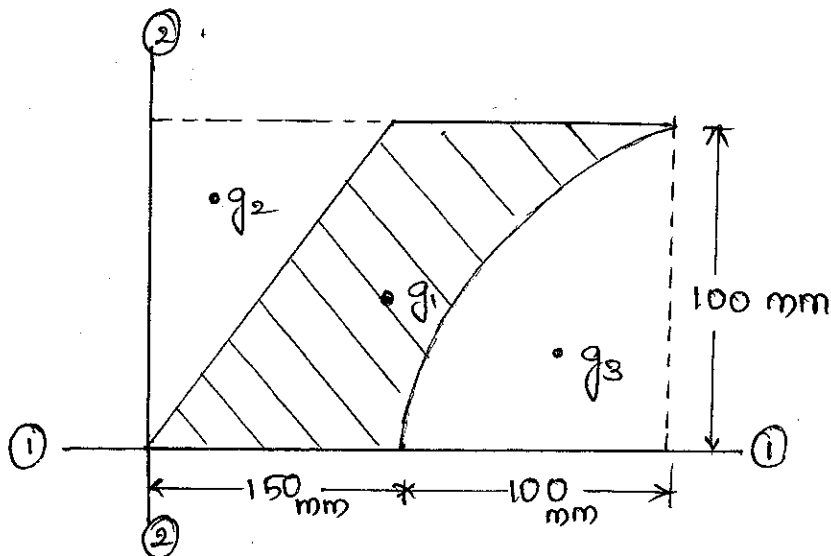
$$I_{xx} = (11.15 \times 10^6) - (5718.2)(39.32)^2$$

$$I_{xx} = 2.3 \times 10^6 \text{ mm}^4$$

$$K_{xx} = \sqrt{\frac{(2.3 \times 10^6)}{5718.2}}$$

$$K_{xx} = 20.1 \text{ mm}$$

8. Find the MI of the region as shown in fig. about horizontal axis (1)-(1) & also find the radius of gyration about the same axis.



Soln:- It is required to find,

$$\bar{y} = \frac{\sum ay}{\sum a}$$

$$I_{1-1} = \sum ay^2 + \sum I_{gx}$$

$$I_{xx} = I_{1-1} - A\bar{y}^2$$

$$K_{xx} = \sqrt{\frac{I_{xx}}{A}}$$

Components	Area (mm ²)	y - mm	ay - mm ³	ay ² - mm ⁴	I _{gx} - mm ⁴
g ₁ (rectangle)	bd 250 × 100 = 25 × 10 ³	$\frac{100}{2}$ = 50	1.25 × 10 ⁶	62.5 × 10 ⁶	$\frac{bd^3}{12} = \frac{250 \times 100^3}{12}$ = 20.83 × 10 ⁶
Deduction g ₂ (triangle)	$\frac{1}{2}bh$ $\frac{1}{2} \times 150 \times 100$ = 7500	$\frac{2}{3}(100)$ = 66.67	500.025 × 10 ³	33.33 × 10 ⁶	$\frac{bh^3}{36} = \frac{150 \times 100^3}{36}$ = 4.16 × 10 ⁶
g ₃ (Quarter circle)	$\frac{\pi r^2}{4} = \frac{\pi(100)^2}{4}$ = 7853.98	$\frac{4r}{3\pi}$ = 42.44	333.32 × 10 ³	14.14 × 10 ⁶	0.05584 = 0.055(100) ⁴ = 5.5 × 10 ⁶
	$\Sigma a =$ 9646.02		$\Sigma ay =$ 416.65 × 10 ³	$\Sigma ay^2 =$ 15.06 × 10 ⁶	$\Sigma I_{gx} =$ 11.17 × 10 ⁶

$$\bar{y} = \frac{416.65 \times 10^3}{9646.02} = \underline{\underline{43.2 \text{ mm}}}$$

$$I_{1-1} = (15.06 \times 10^6) + (11.17 \times 10^6)$$

$$= \underline{\underline{26.23 \times 10^6 \text{ mm}^4}}$$

$$I_{xx} = (26.23 \times 10^6) - (9646.02)(43.2)^2$$

$$I_{xx} = 8.22 \times 10^6 \text{ mm}^4$$

$$K_{xx} = \sqrt{\frac{8.22 \times 10^6}{9646.02}}$$

$$K_{xx} = 29.2 \text{ mm}$$